Electric Power Systems Research 220 (2023) 109352

Contents lists available at ScienceDirect

Electric Power Systems Research

journal homepage: www.elsevier.com/locate/epsr

Load frequency control for power system considering parameters variation using parallel distributed compensator based on Takagi-Sugino fuzzy

Mariem Y. Yousef [a], Magdi A. Mosa [a,*], A.A. Ali [a], A.M.Abdel Ghany [a,b], M.A.Abdel Ghany [c]

[a] Electrical Power and Machines Department, Faculty of Engineering, Helwan University, Cairo, Egypt
[b] Higher Engineering Institute, Thebes academy, Cairo, Egypt
[c] Electrical Power and Machines Engineering Department, Faculty of Engineering, 6 October university, Cairo, Egypt

ARTICLE INFO

March 22, 2023

Keywords:
Polytope
Load frequency control
Takagi-Sugino
Parallel distributed compensator
PID controller
Grey wolf
Tensor product

ABSTRACT

Due to the dependence of large sectors of industrial, commercial, and residential consumers on electricity, the necessity of a reliable and economic generation system that has a regulated frequency is a crucial task. Frequency regulation within permissible standard limits is called load frequency control (LFC). The LFC based on PID controller has proven its effectiveness due to its simple structure and clear concept. However, a fixed gains PID controller can be designed to provide an optimal response at a particular operating condition, but it may exhibit unsatisfactory performance with change in the system's working conditions due to system nonlinearities and parameters variation. To cope with this issue, this article proposes a polytope representation of the system and designs a set of optimal PID controllers at the vertices of the polytope model. After that, Takagi-Sugino fuzzy controller is utilized to blend these optimal controllers to provide the necessary control action based on the actual operating condition. The resultant controller ensures optimal operation at the operation of polytope vertices. To ensure the system's stability over the space of parameter variation, the stability condition as a set of linear matrix inequalities (LMIs) is deduced. Because the number of LMI conditions grows dramatically with the number of system vertices, the paper employs a tensor product transformation technique to represent the system with the minimum number of vertices. The local controllers are optimized using the Grey Wolf Optimizer (GWO) considering a customized objective function that considers the system's integral time absolute error (ITAE), the frequency nadir, and the oscillation. To validate the proposed methodology, the system closed loop poles are determined at a large number of operating conditions. Eventually, the system response is evaluated at different operating conditions with varying load and generation capacity. The proposed control provides a supreme response compared to fixed optimal PID and fixed gain robust PID controller based on H_∞ design criteria controllers.

1. Introduction

The modern power system comprises an interconnection of different generation technologies to provide reliable, secure, and economic generation. To provide a reliable and secure supply without violating the operation limits for generation and load, a balance between generation and demand shall be maintained. The most sensitive variable due to power imbalance is the grid frequency. When the generation becomes greater than the demand, the system's rotating masses accelerate, hence the grid frequency increases, and vice versa [1]. The deviation in the system frequency can harm the generation and the load. Hence, protection readme scenarios are activated, which can cause a cascaded outage and system collapse. Therefore, keeping the grid frequency within the permissible limits is essential, which is usually done through load frequency control (LFC) [1,2]. There are several techniques for LFC, while the PID controller is mostly used in more than 90% of industrial applications due to its simple construction, acceptable performance in many cases, and clear meaning [3]. The challenges and opportunities in LFC are presented in [4] such as requirements of robustness for parameter variation, developing objective functions, and designing optimal controllers.

In [5], design of robust H_2/H_∞ controller for load frequency control of a multi-generation system consisting of an electric vehicle has been introduced. In [6], the load frequency control of two interconnected two

* Corresponding author at: Faculty of Engineering, Helwan University, Cairo, Egypt.
E-mail address: magdimosa@yahoo.com (M.A. Mosa).

https://doi.org/10.1016/j.epsr.2023.109352
Received 31 January 2023; Accepted 18 March 2023
Available online 1 April 2023
0378-7796/© 2023 Elsevier B.V. All rights reserved.

areas has been performed through an optimal PI control concerning the integral time absolute error (ITAE), the controller parameters have been obtained using the sheep fluke optimization technique. Two-degree-of-freedom PID controller has been utilized to provide more flexibility than conventional PID in [7], the controller has five tuning parameters optimized using the sine-cosine technique to optimize the ITAE of frequency and tie-power transfer. In [8], Harris Hawks is used for optimizing the gain of PI controller to optimize the ITAE of the frequency and tie power flow. In [9], robust PI controllers have been designed for multi-area power system using constrained population external optimization through involving the LMIs describing H_∞ performance index into the optimization problem. In [10], Particle Swarm Optimization (PSO) technique has been exploited to optimize the PI controller for controlling super magnetic energy storage (SMES) to regulate the frequency of multi-generation power system with a high penetration of renewable generation resources. In [11], LFC of generation system using PID optimized by using improved version of PSO. In [12], LFC has been implemented by PID for a single area power system which is robust against system measurement noise and parameter variation, however, [11] and [12] do not consider the system nonlinearities and generation rate constraints (GRC). In [13], LFC has been implemented using Integral secondary controller with a state feedback and resilient event generator as an additional control loop to enhance the power system resilience against cyber-physical attack considering the system physical constraints such as GRC. In [14], robust PID controller has been designed for LFC of hybrid power system based on H_∞ design criteria. In [15], salp swarm optimization technique has been utilized for tuning the gains of PID controllers of LFC of multi-area power system. In order to enhance the degree of freedom, fraction order PID (FOPID) control has been proposed to regulate the frequency of a multi-area system in [16], the controller parameters are optimized concerning the ITAE using movable damped wave algorithm (MDVA) optimization technique. In [17], fractional order sliding mode controller with fraction order sliding mode disturbance observer is utilized for LFC of a power system comprises of diesel and wind generation system. In [18], neural network disturbance observer with fractional order integer sliding mode controller (NN-FOISMC) is proposed. In [19], ant lion optimizer has been used to optimize a hybrid PID controller with sliding mode control for the LFC of an interconnected power system. Generally, the sliding mode controllers requires special attention to minimize the control signal chattering. In [20], the LFC of Great Britain's power system, which is considered a single area with a single generation unit, has been implemented using three different control strategies PID, FOPID, and fuzzy PID controllers, which are optimized using the bees' algorithm. In [21], An artificial fuzzy PI, PD, and PID controllers are optimized using artificial bee colony (ABC) for load frequency control of a hybrid power generation system. in [22], hybrid harmony search and cuckoo optimization are utilized to adjust the gains of fuzzy PID control in the load frequency control for a multi-area power system. In [23], fuzzy PID for load frequency control of multi-area power system is tuned using JAYA algorithm. A combination of conventional PI and generalized type 2 fuzzy controller for enhancement of the frequency response has been introduced in [24]. In [25], fractional order PI and fraction order PD have been designed for LFC for single and multi-area power systems. The optimization parameters have been obtained using the dragonfly search technique. In [26], a comparison between PID and model predictive control has been performed using genetic algorithm (GA) and PSO techniques of a two-area power system. In [27], robust PID has been designed to regulate the system frequency with a specific gain margin and phase margin. In [28], parallel distributed PID based on Takagi-Sugino has been introduced for LFC of a single area power system, however, it doesn't guarantee the system's stability in large.

The bulk power system is subjected to an intentional generation outage due to economic operation, or an unintentional outage due to faults. This outage varies the system running capacity and inertia. The LFC shall cope with this challenge. The design of a fixed PID controller may provide an optimal response at a particular operating condition, while the performance may be unsatisfactory in real-time operation [29]. Hence, in this article Parallel distributed PID controller is suggested. The proposed control system consists of a set of optimal controllers and uses Takagi-Sugino fuzzy system to bend the individual controller's output and provide the final control action. In order to consider the system uncertainties, the system is represented by an affine polytope model. The individual PID controllers are optimized at the polytope vertices. The number of vertices depends on the system's nonlinearity, while these number reflects on the required conditions to ensure the system's stability. The linear models of the system can be derived using bisector nonlinearities [30,31]. However, the number of conditions to ensure the system stability becomes very large and reduces the system opportunity. Hence, tensor product is applied to obtain the system polytope representation with a minimum number of vertices and a high degree of accuracy [32,33]. The optimal controllers consider the system nonlinearities such as GRC, and generation limits of the installed units. Grey wolf optimization has proven its capability to optimize nonlinear and nonconvex problems over a large class of metaheuristic techniques [34,35]. Consequently, it is applied to provide optimal controller gains with less time and effort while considering the system's nonlinearity. The objective function is selected to provide a fast-damped response by minimizing the integral time absolute error, frequency overshoot, and oscillation. different constraints such as GRC, upper limits, and lower limits can be involved easily in the design problems. Also, the robust stability conditions to ensure the system ensures the system robust stability over the expected uncertainty space through the satisfaction of a set of linear matrix inequalities which will be developed in the article.

The principal contributions of this article as follows:

1- The proposed PDC-PID controller is able to operate the system optimally at all polytope system vertices, hence it provides a better response compared to fixed controller gains, which operates the system optimally at the tuning point only. Simultaneously, the system stability over the parameter variation space is ensured.
2- Tensor product transformation technique is exploited to develop the fuzzy system membership function to ensure the capability of affine polytope system representation with a minimum number of system vertices
3- Developing the sufficient stability condition of the PDC-PID based on Takagi-Sugino for systems whose control action is a function in the system states, controller states, controller gain, and system parameters using Linear Matrix Inequality technique.
4- Reducing the number of LMI conditions to ensure the system's stability from 512 conditions to 120 conditions by the proper representation of the system. Since, the massive conditions (512) may not be satisfied while the system is stable. The proposed system representation gathers a group of points into a single entity, which provides a significant reduction in the required conditions.
5- Investigation of the closed loop system eigenvalues at randomly selected 486 operating points by introducing an equivalent single PID controller with variable gains for the PDC-PID controller.
6- The performance of the proposed controller is compared with optimal fixed gain PID controller [11] and robust fixed gains PID controller designed using H_∞ criteria [14]

The paper is structured as follows: The first section is the introduction, which illustrated the background of load frequency control, the challenges associated with system parameters variation, and the applied techniques in the literature. Section two elaborates the system modelling by introducing the system block diagram and the transfer function of

each part of the system, which is used to develop the open loop system state space model. After that, a polytope system model is developed to describe the system at different operating conditions using tensor product transformation. Section three presents the operation, structure, and optimal design of the proposed PDC based on the traditional PID and type 3 Takagi-Sugino fuzzy system. In section three, the sufficient conditions for the system's stability are developed. Section four exhibits and analyses the system dynamics using fixed gains PID controllers and the proposed PDC. Section five develops an equivalent single PID controller for the PDC with variable gains and illustrates the location of the system eigenvalues in the S-plan over a wide range of system variables. At the end of the article, in section six, the findings made during the research are summarized.

2. Power system modelling

The Egyptian power system has a variety of generation plant technologies, that are strongly coupled. For load frequency control the national energy control centre (NECC), of the Egyptian Electricity Authority, has introduced a dynamic model of the EPS [28]. This model comprises three technologies, which are hydraulic plant stations, thermal non-reheat stations, and thermal reheat stations. The details and corresponding state space model are discussed in this section as follows.

2.1. System block diagram

In this subsection, the transfer function of each part of the proposed power system is introduced and the overall system block diagram is displayed.

2.1.1. Swing equation

The swing equation that governs the frequency variation of the Egyptian power system is given by Eq. (1) [2]. Where:Δf is the power system frequency deviation, ΔP_{m1}, reheat ΔP_{m2}, and hydraulic stations ΔP_{m3} $\Delta P_m$is the total mechanical power input to the grid from heat, reheat, and hydraulic units. ΔP_L is the load power. *H and D* are the inertia and damping factors of the system, respectively.

$$\Delta f = \frac{1}{2Hs + D}(\Delta P_{m1} + \Delta P_{m2} + \Delta P_{m3} - \Delta P_L) \tag{1}$$

2.1.2. Non-reheat station

The mechanical power of the reheat plant is given by Eq. (2), where: ΔP_{c1} is the governor's set point, R_1 is the governor regulation factor, T_1 is the governor's time constant, and P_{n1} is the per unit nominal working capacity of the non-reheat plant.

$$\Delta P_{m1} = \frac{P_{n1}}{T_1 s + 1}\left(-\frac{1}{R_1}\Delta f - \Delta P_{c1}\right) \tag{2}$$

2.1.3. Reheat station

Governor droop control of the reheat plant is given by Eq. (3), where ΔP_{c2} is the governor's set point, R_1 is the governor regulation factor, T_2 the governor's time constant, and P_{n2} is the per unit nominal working capacity of the reheat plant.

$$\Delta P_{g2} = \frac{P_{n2}}{T_2 s + 1}\left(-\frac{1}{R_2}\Delta f - \Delta P_{c2}\right) \tag{3}$$

The reheat power plant consists of a multi (high pressure, intermediate pressure, and low pressure) steam turbine with a reheater after the high-pressure stage. The output power of the high-pressure stage is a fraction of the governor mechanical power as given by Eq. (4), where m is the fraction factor. The remaining power is obtained after reheating and passing on intermediate and low-pressure stages as given in Eq. (5), where T_b is the reheater time constant. The mechanical power of the reheater power plant is the summation of high-pressure output and output after reheating as shown in Eq. (6).

$$\Delta P_{m2_hp} = m\Delta P_{g2} \tag{4}$$

$$\Delta P_{T_lp_Lp} = \frac{1-m}{T_b s + 1}\Delta P_{g2} \tag{5}$$

$$\Delta P_{m2} = \Delta P_{m2_hp} + \Delta P_{m2_lp_Lp} = \frac{mT_b s + 1}{T_b s + 1}\Delta P_{g2} \tag{6}$$

2.1.4. Hydraulic station

Governor droop control of the hydraulic station is given by Eq. (7). Where: T_d is The hydro governor dynamics including the dashpot time, T_3 is the governor's time constant, R_3 is the governor regulation factor, ΔP_{c2} is the governor's set point, and P_{n3} is the per unit nominal working capacity of the hydraulic plant. The output power of the hydraulic station is given by the turbine-penstock model as given by Eq. (8) [36], T_W is water time constant or water starting time.

$$\Delta P_{g3} = P_{n3}\frac{T_d s + 1}{T_3 s + 1}\left(-\frac{1}{R_3}\Delta f - \Delta P_{c3}\right) \tag{7}$$

$$\Delta P_{m3} = \frac{-T_w s + 1}{0.5T_W s + 1}\Delta P_{g3} \tag{8}$$

The block diagram of the EPS is given in Fig. 1. the model includes the system nonlinearity in terms of saturation and GRC. The GRC is 0.1 pu/min, 0.2 pu/min, and 0.5 pu/min for reheat stations, non-reheat stations, and hydraulic stations respectively. the remaining parameters are given as follows: T_1=T_2=0.4 s, T_3=90 s, T_b=6 s, T_w =1 s,T_d =5 s., R_1 =2.5 Hz/ p.u, R_2 =2.5 Hz/ p.u, R_3=1 Hz/ p.u MW, D= 0.028 p.u /Hz, and m = 0.5.

2.2. State space modelling of the system

The open loop control system, without the controller, block diagram of Fig. 1 is a six-order system, which can be described in state space using seven state variables. The state variables are selected as given in (9), the control input is the governor set point as given by Eq. (10), and the disturbance input is the load as given by Eq. (11). The output is the frequency deviation as given by Eq. (12). The state space model in standard form is given by Eqs. (13) and (14). Where: $x_P(t), u(t),\ W(t),$ *abd* $y(t)$ are the open loop system state vector, input vector, disturbance vector and output vector. A_P, B_P and E_p are the open loop state matrix, input matrix, and disturbance matrix. *C and D* are the output matrix and transfer matrix.

$$x_p(t) = [\,\Delta f(t) \quad \Delta P_{m1}(t) \quad \Delta P_{m2}(t) \quad \Delta P_{g2}(t) \quad \Delta P_{m3}(t) \quad \Delta P_{g3}(t)\,]^T \tag{9}$$

$$u(t) = \Delta P_{c1}(t) = \Delta P_{c2}(t) = \Delta P_{c3}(t) \tag{10}$$

$$W(t) = \Delta P_L(t) \tag{11}$$

$$y(t) = \Delta f(t) \tag{12}$$

$$\dot{x}_p(t) = A_P x_P(t) + B_P u(t) + E_p W(t) \tag{13}$$

$$y(t) = Cx_P(t) + Du(t) \tag{14}$$

Where:

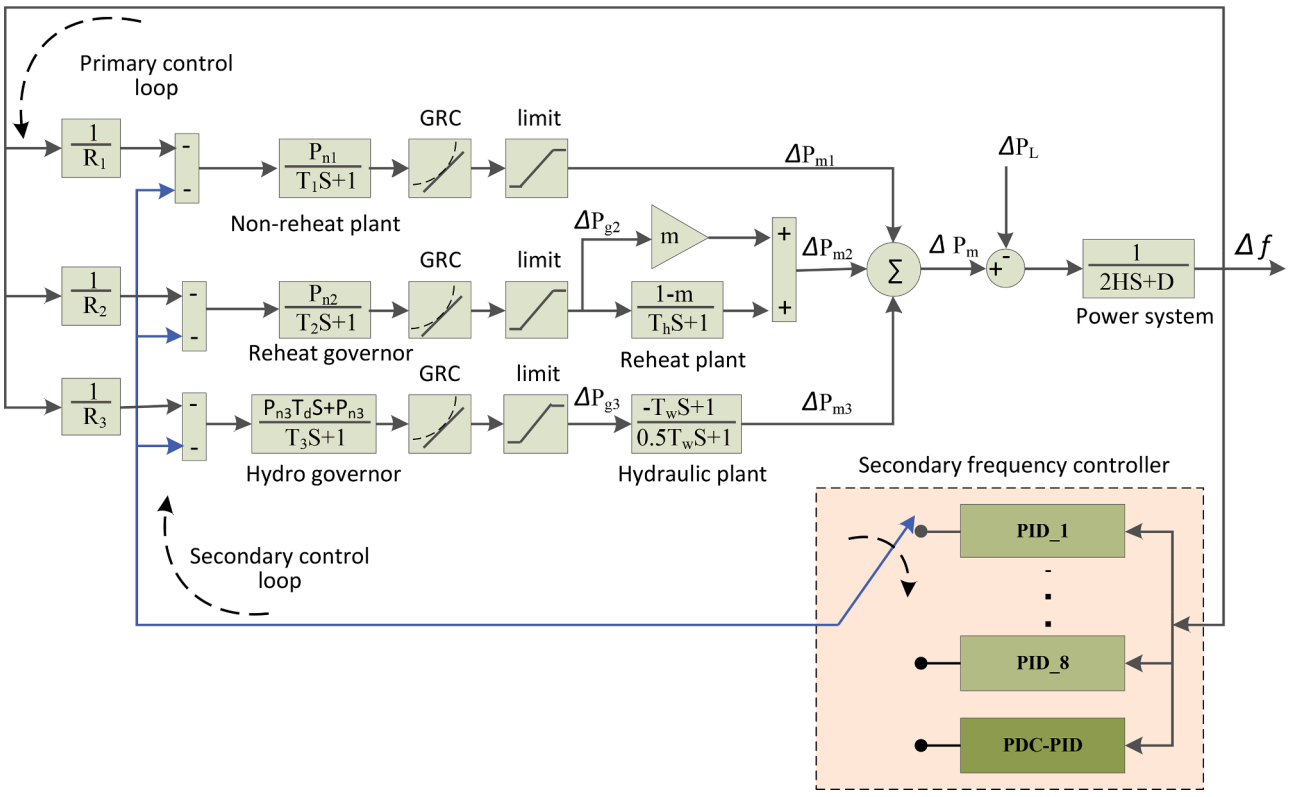


Fig. 1. Block diagram of the Egyptian power system.

$$A_P = \begin{bmatrix} -\frac{D}{2H} & \frac{1}{2H} & \frac{1}{2H} & 0 & \frac{1}{2H} & 0 \\ -a_1 & -\frac{1}{T_1} & 0 & 0 & 0 & 0 \\ -ma_2 & 0 & -\frac{1}{T_h} & b_1 & 0 & 0 \\ -a_2 & 0 & 0 & -\frac{1}{T_2} & 0 & 0 \\ -2(Db_2 - a_3) & 2b_2 & 2b_2 & 0 & \left(2b_2 - \frac{2}{T_W}\right) & \left(\frac{2}{T_3} + \frac{2}{T_W}\right) \\ (Db_2 - a_3) & -b_2 & -b_2 & 0 & -b_2 & -\frac{1}{T_3} \end{bmatrix} \quad (15)$$

$$B_P = \begin{bmatrix} 0 & -\frac{P_{n1}(t)}{T_1} & -\frac{mP_{n2}(t)}{T_2} & -\frac{P_{n2}(t)}{T_2} & \frac{2P_{n3}(t)}{T_3} & -\frac{P_{n3}(t)}{T_3} \end{bmatrix}^T \quad (16)$$

$$E = \begin{bmatrix} -\frac{1}{2H} & 0 & 0 & 0 & -2b_2 & b_2 \end{bmatrix}^T$$

$$C = [1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0] \text{ and } D = [0]$$

$$a_1 = \frac{P_{n1}(t)}{R_1 T_1},\ a_2 = \frac{P_{n2}(t)}{R_2 T_2},\ a_3 = \frac{P_{n3}(t)}{R_3 T_3},$$
$$a_4 = \frac{T_h}{2H},\ b_1 = \frac{1}{T_h} - \frac{m}{T_2},$$
$$b_2 = a_3 a_4 = \frac{P_{n3}}{R_3 T_3}\frac{T_h}{2H}$$

2.3. Polytope representation of the open loop system

The open loop control system has variable parameters $P_{n1}(t)$, $P_{n2}(t)$, and $H(t)$. While $P_{n3}(t)$ is a linear combination of $P_{n1}(t)$ and $P_{n2}(t)$. Therefore, this system is linear with parameter variation and is called a linear parameter varying (LPV) system as given by Eq. (13), which can be rewritten in the form of Eq. (17). Where $p(t)$ is the variable parameters. the range of parameter variations is given by inequalities (19)-(21). The bounded space of parameter variation is represented by a cube; whose vertices are the extreme values of variable parameters as shown in Fig. 2.

$$\dot{x}_P(t) = A_P(p(t))x_P(t) + B_P(p(t))u + E_p(p(t))W(t) \quad (17)$$

$$p(t) = f(H(t),\ p_{n1}(t), p_{n2}(t)) \quad (18)$$

$$0.1 \le P_{n1}(t) \le 0.35 \quad (19)$$

$$0.2 \le P_{n2}(t) \le 0.64 \quad (20)$$

$$4 \le H(t) \le 8 \quad (21)$$

Analysis and control design of parameters varying system is not an easier task, one of the suitable techniques that can handle this

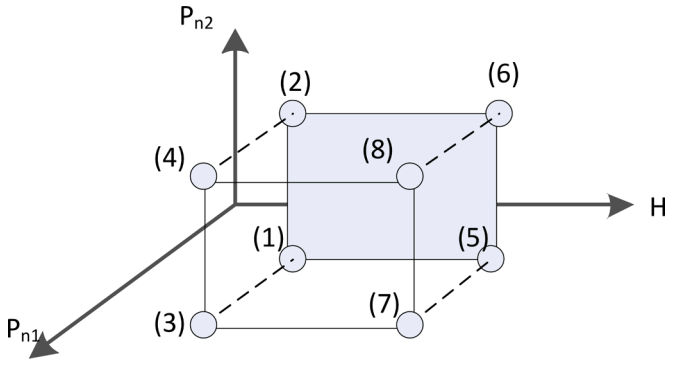


Fig. 2. Bounded space of parameter varying.

Table 1
variables at different vertices.

Operating point (Vertex)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
H	4	4	4	4	8	8	8	8
P_{n1}	0.1	0.1	0.35	0.35	0.1	0.1	0.35	0.35
P_{n2}	0.2	0.64	0.2	0.64	0.2	0.64	0.2	0.64

complexity is modeling the system at a group of operating points called vertices. These points are selected in such a way that the system can be represented as a linear combination of the selected vertices. This system representation is called polytope model [37]. The number of vertices depends on the number of dependent variables and the number of nonlinear elements. The minimum number of vertices is equals 2^N, where N is the number of independent variables, in our case $N = 3$. Hence, the minimum number of polytope vertices is 8, corresponding to the extreme values of variable parameters as shown in Fig. 2. The location of polytope vertices in the variable space is summarized in table 1.

The system can be represented as a linear combination of the polytope vertices as given by Eq. (22) and (23). Where W_r is the weighting of the vertex r. The individual weight of each vertex shall satisfy condition (24). At the same time, the summation of all weights shall equal unity as described in Eq. (25).

$$A_P(p(t)) = W_1(p(t))A_1 + W_2(p(t))A_2 + \ldots + W_R(p(t))A_R = \sum_{r=1}^{R} W_r(p(t))A_r \tag{22}$$

$$B_P(p(t)) = W_1(p(t))B_1 + W_2(p(t))B_2 + \ldots + W_R(p(t))B_R = \sum_{r=1}^{R} W_r(p(t))B_r \tag{23}$$

$$\forall\ r \in [1, R],\ p(t):\ 0 \le\ W_r(p(t)) \le 1 \tag{24}$$

$$\forall\ p(t):\ \sum_{r=1}^{R} W_r(p(t)) = 1 \tag{25}$$

The polytope model can be represented by Takagi-Sugino fuzzy type 3. Since the number of vertices is the number of rules and the linear combination is the summation of the individual fuzzy output. Hence, the polytope in form of fuzzy can be rewritten as given by Eq. (26) and (27), or more compact in form of (28) and (29). Where, $W_r(p(t))$ is the weight of rule r. $W_{1,i}(p_1(t))$, $W_{2,j}(p_2(t))$, *and* $W_{3,k}(p_3(t))$ are the grade of variables $p_{n1}(t)$, $p_{n2}(t)$, *and* $H(t)$in membership function $(i, j,$ *and* $k)$ respectively.

$$\begin{aligned} A_P(p(t)) \approx\ & W_{1,1}(p_1(t))W_{2,1}(p_2(t))W_{3,1}(p_3(t))A_{1,1,1} \\ & + W_{1,1}(p_1(t))W_{2,1}(p_2(t))W_{3,2}(p_3(t))A_{1,1,2}\ + \ldots \\ & +\ W_{1,2}(p_1(t))W_{2,2}(p_2(t))W_{3,2}(p_3(t))A_{2,2,2} \end{aligned} \tag{26}$$

$$\begin{aligned} B_P(p(t)) \approx\ & W_{1,1}(p(t))W_{2,1}(p_2(t))W_{3,1}(p_3(t))B_{1,1,1} \\ & + W_{1,1}(p_1(t))W_{2,1}(p_2(t))W_{3,2}(p_3(t))B_{1,1,2}\ + \ldots \\ & +\ W_{1,2}(p_1(t))W_{2,2}(p_2(t))W_{3,2}(p_3(t))B_{2,2,2} \end{aligned} \tag{27}$$

$$\begin{aligned} A_P(p(t)) &\approx \sum_{r=1}^{R} W_r(p(t))A_r \\ &\approx \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} W_{1,i}(p_1(t))W_{2,j}(p_2(t))W_{3,k}(p_3(t))A_{i,j,k} \end{aligned} \tag{28}$$

$$\begin{aligned} B_P(p(t)) &\approx \sum_{r=1}^{R} W_r(p(t))B_r \\ &\approx \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} W_{1,i}(p_1(t))W_{2,j}(p_2(t))W_{3,k}(p_3(t))B_{i,j,k} \end{aligned} \tag{29}$$

For exact polytope system representation, the sector nonlinearity technique can be used. However, the sector non-linearity produces a large number of system vertices which makes the system analysis for stability complex. Therefore, the tensor product technique is utilized to determine the membership functions which minimizes the error ε between the system's actual model and the linear combination model as given by Eq. (30). The optimization statement is given by Eq.(31). The details of tensor product transformation can be found in [32,33]. The resultant membership functions are shown in Fig. 3.

$$\begin{aligned} &\| \left[A_P(p(t)) \quad B_P(p(t)) \right] - \sum_{i=1}^{I}\sum_{j=1}^{J} \\ &\times \sum_{k=1}^{K} W_{1,i}(p_1(t))W_{2,j}(p_2(t))W_{3,k}(p_3(t))[A_{i,j,k} \quad B_{i,j,k}]\| \le \varepsilon \end{aligned} \tag{30}$$

Polytope optimization problem:

$$\begin{aligned} &\min\ \varepsilon \\ &W_{1,i}(p_1(t))W_{2,j}(p_2(t))W_{3,j}(p_3(t))\ \forall\ i \in I,\ j \in J,\ k \in K \\ &s.t.\ \textit{inequality}\ (30) \\ &\textit{inequality}\ (24) \\ &\textit{equality}\ (25) \\ &\textit{inequality}\ (19)\ \textit{to}\ (21) \end{aligned} \tag{31}$$

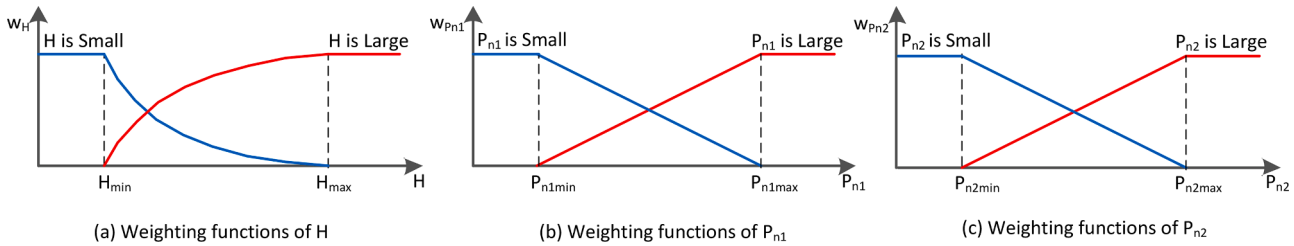
In order to check the validity of polytope model representation using suggested weighting functions and eight vertices for cubic representation, the error between polytope representation and actual model representation is obtained at assumed 486 different operating conditions for variation of inertia, the nominal power of non-reheat and nominal power of reheat by 0.5, 0.05 and 0.05 respectively. The percentage of the maximum error inside the system matrix is zero at $H = 4$ and $H = 8$ sec. as shown in Fig. 4 irrespective of the value of $P_{n1}$and P_{n2}. While at other values of H, the maximum error is less than 0.002 % per unit, which is very small and can be neglected. This tiny error is because of the curvature approximation of the inertia weighting function as shown in Fig. 3(a).

3. Parallel compensator PID proposed controller

In this section, the structure and operation of the proposed fuzzy controller are elaborated. After that, the necessary stability condition is developed. In the end, the optimal controller is designed at polytope vertices.

3.1. Structure and operation of PDC

The power system may be subjected to an outage of generation units due to abnormal conditions such as faults, scheduled maintenance, or economic operation. This outage affects the generation capacity of the power system in terms of the nominal working capacity of the units. In addition to the change in the rotating inertia of the power system. These changes in operation effects the system's dynamic performance. Hence, the design of an optimal PID controller at a particular operating point may provide an unstable or unsatisfactory response at another operating point as will be seen in the simulation. Therefore, multi-controllers can be used since each controller is optimal at a single operating point. As Takagi-Sugino fuzzy type 3 controller can be used to produce a nonlinear model from a set of linear models with a smooth transition



(a) Weighting functions of H (b) Weighting functions of P_{n1} (c) Weighting functions of P_{n2}

Fig. 3. Membership (Weighting) functions of different variables.

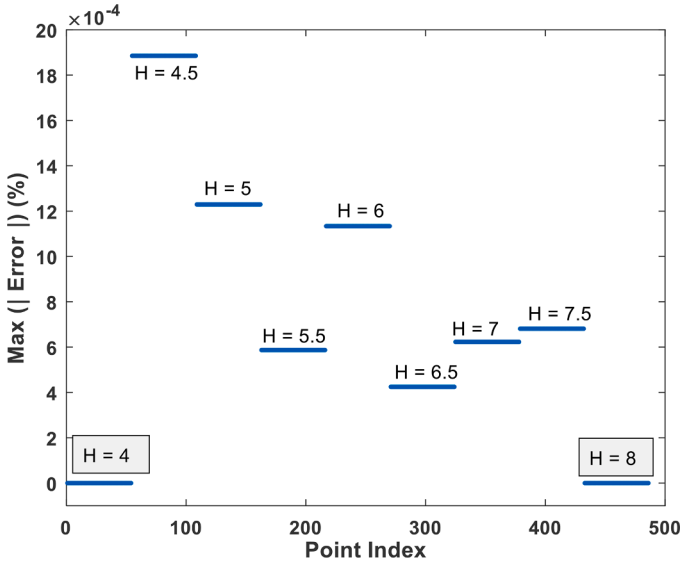


Fig. 4. Maximum error between Polytopic model using tensor product and exact system model.

Table 2
Rule base of PDCPID.

$H(t)$ is Small			$H(t)$ is Large		
$p_{n2}(t)$	$p_{n1}(t)$		$p_{n2}(t)$	$p_{n1}(t)$	
	Small	Large		Small	Large
Small	PID_1	PID_3	Small	PID_5	PID_7
Large	PID_2	PID_4	Large	PID_6	PID_8

from one point to another point. Hence it is utilized in this paper. The navigation and combination of the controllers are achieved according to the set of the input variable. The design starts with defuzzification since each input is split into two fuzzy sets at least as shown in Fig. 2. These membership functions are obtained using tensor product technique as discussed early in the previous section. the second stage is the rule base, since we have three variables each has two membership functions, then we have eight rules. The summary of the rule base is given in Table 2. The consequent part of Takagi-Sugino rule is a dynamic module which is the corresponding controller as shown in Table 3. The individual rule controller is represented in terms of Eq. (32). The output of the PDCPID is the weighted average of the output of each controller according to the rule base weight as given in Eq. (33). Where R is the number of rules

Table 3
Takagi-Sugino code.

Rule#1
If H is small and $p_{n1}(t)$ *is small and* $p_{n2}(t)$ *is small, then*
$u_1(t) = K_{p1}\Delta f(t) + K_{i1}\int \Delta f(t)dt + K_{d1}\frac{d\Delta f(t)}{dt}$.
.
.
Rule #8
If H is Large and $p_{n1}(t)$ *is Large and* $p_{n2}(t)$ *is large, then*
$u_8(t) = K_{p8}\Delta f(t) + K_{i8}\int \Delta f(t)dt + K_{d8}\frac{d\Delta f(t)}{dt}$

(system vertices). A simplified schematic diagram of the proposed PDC is shown in Fig. 5.

$$u_j(t) = K_{pj}\Delta f(t) + K_{ij}\int \Delta f(t)dt + K_{dj}\frac{d\Delta f(t)}{dt} \tag{32}$$

$$u(t) = \frac{\sum_{r=1}^{R} W_r u_r}{\sum_{r=1}^{R} W_r} = \sum_{r=1}^{R} W_r u_r \tag{33}$$

3.2. Sufficient stability condition of the proposed control system

The stability of the closed control system is an intuitive requirement that shall be achieved. The presence of parameter variation in the system model in addition to utilizing non-linear controllers requires additional effort to prove the system's stability. In this section sufficient condition of the closed system stability using the proposed PDC is developed and checked. Before, proceeding recall that the seed of the controller is the PID controller which is a dynamic controller due to the presence of the integral part as shown in Eq. (32). Hence, the system states increase by one. The new state is defined by Eq. (34) corresponding to the first derivative is given by Eq. (35), where $x_c(t)$ is the compensator state. To make the derivation simple, the controller state is embedded into the open loop system states as given by Eq. (36). Where: $x(t)$ is the system states. $A(p(t))$ *and* $B(p(t))$, the state matrix and input matrix after adding the controller state.

$$x_c(t) = \int \Delta f(t)dt \tag{34}$$

$$\dot{x}_c(t) = \Delta f(t) = x_1(t) \tag{35}$$

$$\dot{x}(t) = A(p(t))x(t) + B(p(t))u(t) + F(t)W(t) \tag{36}$$

$$x(t) = \begin{bmatrix} x_P(t) & x_c(t) \end{bmatrix}^T = \begin{bmatrix} \Delta f(t) & \Delta P_{m1}(t) & \Delta P_{m2}(t) & \Delta P_{g2}(t) & \Delta P_{m3}(t) & \Delta P_{g3}(t) & \int \Delta f(t)dt \end{bmatrix}^T \tag{37}$$

$$A(p(t))=\begin{bmatrix} -\frac{D}{2H} & \frac{1}{2H} & \frac{1}{2H} & 0 & \frac{1}{2H} & 0 & 0 \\ -a_1 & -\frac{1}{T_1} & 0 & 0 & 0 & 0 & 0 \\ -ma_2 & 0 & -\frac{1}{T_h} & b_1 & 0 & 0 & 0 \\ -a_2 & 0 & 0 & -\frac{1}{T_2} & 0 & 0 & 0 \\ -2(Db_2-a_3) & 2b_2 & 2b_2 & 0 & \left(2b_2-\frac{2}{T_w}\right) & \left(\frac{2}{T_3}+\frac{2}{T_w}\right) & 0 \\ (Db_2-a_3) & -b_2 & -b_2 & 0 & -b_2 & -\frac{1}{T_3} & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{38}$$

$$B(p(t))=\begin{bmatrix} 0 & -\frac{P_{n1}}{T_1} & -\frac{mP_{n2}}{T_2} & -\frac{P_{n2}}{T_2} & \frac{2P_{n3}}{T_3} & -\frac{P_{n3}}{T_3} & 0 \end{bmatrix}^T \tag{39}$$

It should be noted that the A and B matrices of the system (36) are the same as the open loop system Eq. (15) and (16) after adding a constant column and raw due to the new compensator state. Therefore, the new system is a polytope with eight vertices as the open loop system. Hence, the system can be represented using polytope form as indicated by Eq. (40), where R is the number of polytope vertices or the number of fuzzy rules.

$$\dot{x}(t)=\sum_{i=1}^{R} h_i(p(t))A_i x(t)+\sum_{i=1}^{R} h_i(p(t))B_i u(t) \tag{40}$$

The input control signal is the output of the PDC controller, the output of each rule of PDC is given by the PID Eq. (32). Substitute the system frequency derivative from Eq. (13) to obtain the controller output as a function in the system state as shown in Eq. (41)

$$u_j(t)=K_{pj}\Delta f(t)+K_{ij}\int \Delta f(t)dt+K_{dj}\Big(-\frac{D}{2H}\Delta f(t)+\frac{1}{2H}\Delta P_{m1}(t)+\frac{1}{2H}\Delta P_{m2}(t)+\frac{1}{2H}\Delta P_{m3}(t)\Big) \tag{41}$$

Eq. (41) shows that the controller output has two parts, first part depends on the controller gains (K_p *and* K_i) and the second part depends on both K_d *and* H. Hence, the input is divided into two signals (u_{1j} and u_{2j}) as shown in Eq. (42) and (43).

$$u_{1j}(t)=K_{pj}\Delta f(t)+K_{ij}\int \Delta f(t)dt=F_j x(t) \tag{42}$$

$$u_{2j}(t)=K_{dj}\Big(-\frac{D}{2H}\Delta f+\frac{1}{2H}\Delta P_{m1}+\frac{1}{2H}\Delta P_{m2}+\frac{1}{2H}\Delta P_{m3}\Big)=K_{dj}Zx(t) \tag{43}$$

$$F_j=[K_{pj}\quad 0\quad 0\quad 0\quad 0\quad 0\quad K_i j] \tag{44}$$

$$Z=\begin{bmatrix} -\frac{D}{2H} & \frac{1}{2H} & \frac{1}{2H} & 0 & \frac{1}{2H} & 0 & 0 \end{bmatrix} \tag{45}$$

The vector Zdepends on the system parameters, if the open loop system is modeled in polytope then the vector Z_k at each vertex of the polytope as shown in Eq. (46). The operating Z vector is calculated from the vertices as given by Eq. (47), where M_k is the weighting of vertex k and $\sum_{k=1}^{r} M_k = 1$. Substitute from Eq. (47) into Eq. (43). The second signal of input is obtained as shown in Eq. (48).

$$Z_k=\begin{bmatrix} -\frac{D}{2H_k} & \frac{1}{2H_k} & \frac{1}{2H_k} & 0 & \frac{1}{2H_k} & 0 & 0 \end{bmatrix} \tag{46}$$

$$Z=\sum_{k=1}^{R} M_k Z_k \tag{47}$$

$$u_{2j}=K_{dj}\sum_{i=1}^{r} M_k Z_k x(t) \tag{48}$$

The output of all rules of the PDC is the summation of each output rule as given by Eq. (49). Where W_j is the weighting of the rule or the controller j. Substitute from Eq. (48) and Eq. (42) into Eq. (49). The final output of PDC is obtained as indicated by Eq. (50). Which can be rewritten in form of Eq. (51).

$$u(t)=\sum_{j=1}^{R} W_j u_{1j}(t)+\sum_{j=1}^{R} W_j u_{2j}(t) \tag{49}$$

$$u(t)=\sum_{j=1}^{R} W_j F_j x(t)+\sum_{j=1}^{R} W_j K_{dj}\sum_{k=1}^{R} M_k Z_k x(t) \tag{50}$$

$$u(t)=\sum_{j=1}^{R} W_j F_j x(t)+\sum_{j=1}^{R}\sum_{k=1}^{R} W_j M_k K_{dj} Z_k x(t) \tag{51}$$

Substitute the input signal from Eq. (51) into Eq. (40), hence the system dynamics is described by Eq. (52). The state equation is rearranged in form of Eq. (53) and (54).

$$\dot{x}(t)=\sum_{i=1}^{R} h_i A_i x(t)+\sum_{i=1}^{R} h_i B_i\left(\sum_{j=1}^{R} W_j F_j x(t)+\sum_{j=1}^{R}\sum_{k=1}^{R} W_j M_k K_{dj} Z_k x(t)\right) \tag{52}$$

$$\dot{x}(t)=\sum_{i=1}^{R} h_i A_i x(t)+\sum_{i=1}^{R} h_i B_i\sum_{j=1}^{R} W_j F_j x(t)+\sum_{i=1}^{r} h_i B_i\sum_{j=1}^{r}\sum_{k=1}^{r} W_j M_k K_{dj} Z_k x(t) \tag{53}$$

$$\dot{x}(t)=\sum_{i=1}^{R} h_i A_i x(t)+\sum_{i=1}^{R}\sum_{j=1}^{R} h_i W_j B_i F_j x(t)+\sum_{i=1}^{R}\sum_{j=1}^{R}\sum_{k=1}^{R} h_i W_j M_k B_i K_{dj} Z_k x(t) \tag{54}$$

According to affine polytope representations, the equalities (55) are valid. Hence the first term of Eq. (54) can be written in form of Eq. (56). Similarly, the second term of Eq. (54) can be written as given by Eq. (57).

$$\sum_{i=1}^{R} h_i=1,\ \sum_{j=1}^{R} W_j=1\ and\ \sum_{k=1}^{R} M_k=1 \tag{55}$$

$$\sum_{i=1}^{R} h_i A_i=\sum_{j=1}^{R} W_j\left(\sum_{i=1}^{R} h_i A_i\right)=\sum_{i=1}^{R}\sum_{j=1}^{R} h_i W_j A_i=\sum_{k=1}^{R} M_k\left(\sum_{i=1}^{R}\sum_{j=1}^{R} h_i W_j A_i\right)=\sum_{i=1}^{R}\sum_{j=1}^{R}\sum_{k=1}^{R} h_i W_j M_k A_i \tag{56}$$

$$\sum_{i=1}^{R}\sum_{j=1}^{R} h_i W_j B_i F_j=\sum_{i=1}^{R}\sum_{j=1}^{R}\sum_{k=1}^{R} h_i W_j M_k B_i F_j \tag{57}$$

Substitute from Eq. (56) and (57) into Eq. (54), the system closed-loop state equation is obtained as shown in Eq. (58). Which is rearranged inform of Eq. (59).

$$\dot{x}(t) = \sum_{i=1}^{R}\sum_{j=1}^{R}\sum_{k=1}^{R} h_i W_j M_k A_i x(t) + \sum_{i=1}^{R}\sum_{j=1}^{R}\sum_{k=1}^{R} h_i W_j M_k B_i F_j x(t) + \sum_{i=1}^{R}\sum_{j=1}^{R} \times \sum_{k=1}^{R} h_i W_j M_k B_i K_{dj} Z_k x(t) \tag{58}$$

$$\dot{x}(t) = \sum_{i=1}^{R}\sum_{j=1}^{R}\sum_{k=1}^{R} h_i W_j M_k \left(A_i + B_i F_j + B_i K_{dj} Z_k\right) x(t) \tag{59}$$

Eq. (59) is a polytope representation of the closed loop system, hence according to Lyapunov stability, the system is stable if there is a common positive definite symmetrical matrix $S > 0$, satisfying the inequality (50) for all possible system vertices.

$$\left(A_i + B_i F_j + B_i K_{dj} Z_k\right)^T S + S\left(A_i + B_i F_j + B_i K_{dj} Z_k\right) \langle 0 \quad \forall\ i,j,k\ \in \{1,2,\cdots,\ R\} \tag{60}$$

The Lyapunov matrix S has to satisfy all the inequalities conditions of the form (50), the number of Lyapunov inequalities equals to R^3. In this paper the number of rules is 8 rules, hence the number of inequality conditions is 512 which is a massive number of conditions. Increasing the number of conditions may lead to stalling of LMI solvers due to slower progress. In the proposed system, if the system bases are used similarly to the controller bases (i.e. the same membership functions and the same rule premise), Eq. (61) is valid. This feature can be utilized to reduce the number of required conditions. According to Eq. (61), the closed loop system of Eq. (59) can be written in form of Eq. (63), System dynamic using Eq. (63) is the same as Eq. (59), while Eq. (63) has less number of closed loop system vertices.

$$h_i = W_j = M_k \ \forall\ i = j = k\ \in \{1,2,\ldots,\ R\} \tag{61}$$

The number of LMIs for stability conditions (63)-(65) is $\frac{R(R+1)(R+2)}{6}$. For $R = 8$, the number of LMIs is 120. Hence, the system representation using (62) has more relaxed conditions compared to representation in form of (59).

3.3. Optimal PID Controller at different points

The objective function at each operating point is selected to compromise between the frequency nadir, oscillation, and restoration time as shown in Eq. (66). The first term considers the frequency nadir, the second term enhances the system damping by minimizing frequency oscillation, and the third term is the integral time absolute error (ITAE) to reduce the system restoration time. The constants $k_1,\ k_2,\ and\ k_3$ are weighting factors, and can act as unit conversion.

$$OF = k_1 |\Delta f(t)|_{max} + k_2 \max(\Delta f(t)) + k_3 \int_0^t t|\Delta f(t)|\ dt \tag{66}$$

There are numerous metaheuristic techniques, PSO and GWO optimization prove their ability to optimize a problem of a higher degree of nonlinearity and complexity. The GWO proved it capability of solving complex and engineering optimization problems compared to many metaheuristic techniques [29]. In addition, the GWO is able to provide optimal fixed gains PID controller for LFC in [38,39]. In addition, GWO structure rule enhances the algorithm capability of exploring the region of the optimal solution and hierarchically to determine the optimal solution within the region. Hence, GWO is recommended in this article. The GWO is imitating the hunting strategy of grey wolf species. The grey wolves live in a group, which has a restricted structure, where the individuals in the pack are categorized into four categories known as

$$\dot{x}(t) = \sum_{i=1}^{R} h_i W_j M_k G_{iii} x(t) + \sum_{i=1}^{R}\sum_{k=1+1}^{R}\sum_{j=k+1}^{R} 6h_i W_j M_k \left(\frac{G_{ijk} + G_{ikj} + G_{jik} + G_{jki} + G_{kji} + G_{kij}}{6}\right) x(t) + \sum_{\substack{i=1\\k=i}}^{v}\sum_{j\neq 1}^{v} 3h_i W_j M_i \left(\frac{G_{iji} + G_{iij} + G_{jii}}{3}\right) x(t) \tag{62}$$

Where: $G_{ijk} = A_i + B_i F_j + B_i K_{dj} Z_k$

Recall, the system in form (62), is stable in large if there is a common positive definite symmetrical matrix $S > 0$ satisfies the Lyapunov inequality at each of the system vertices. Hence, the system is stable in large if there is a common symmetrical matrix $S > 0$ that satisfies the conditions (63)-(65).

$$G_{i,i,i}^T S + S G_{i,i,i} < 0\ \forall i \in \{1,2,\ldots,\ R\} \tag{63}$$

alpha, beta, delta, and omega as shown in Fig. 6. The top of the group is the alpha category, which is the leader of the group and it is responsible for deciding the hunting, sleeping place, wake time, and so on. The alpha is not necessarily the strongest member of the group however; it is the best term for managing the group. In problem optimization, the fittest solution is considered the alpha wolf. The second category is denoted as beta, which is the best candidate to be alpha when the alpha died or becomes very older. It acts as an advisor for the alpha and a disciplinary for the rest of the wolf pack. In the optimization technique, the second-ranked fitted solution is selected as the beta wolf. The third category is the delta which acts as scouts, caretakers, and hunters. In

$$\left(\frac{G_{i,j,k} + G_{i,k,j} + G_{j,i,k} + G_{j,k,i} + G_{k,j,i} + G_{k,i,j}}{6}\right)^T S + S\left(\frac{G_{i,j,k} + G_{i,k,j} + G_{j,i,k} + G_{j,k,i} + G_{k,j,i} + G_{k,i,j}}{6}\right) < 0 \quad \forall i,j,k \in \{1,2,\cdots,\ R-1\},\ k\rangle i,\ j > k \tag{64}$$

$$\left(\frac{G_{i,j,i} + G_{j,i,i} + G_{i,i,j}}{3}\right)^T S + S\left(\frac{G_{i,j,i} + G_{j,i,i} + G_{i,i,j}}{3}\right) < 0 \quad \forall i \in \{1,2,\cdots,\ R-1\},\ j \in \{1,\cdots,\ R-1,R\}\ -\{i\} \tag{65}$$

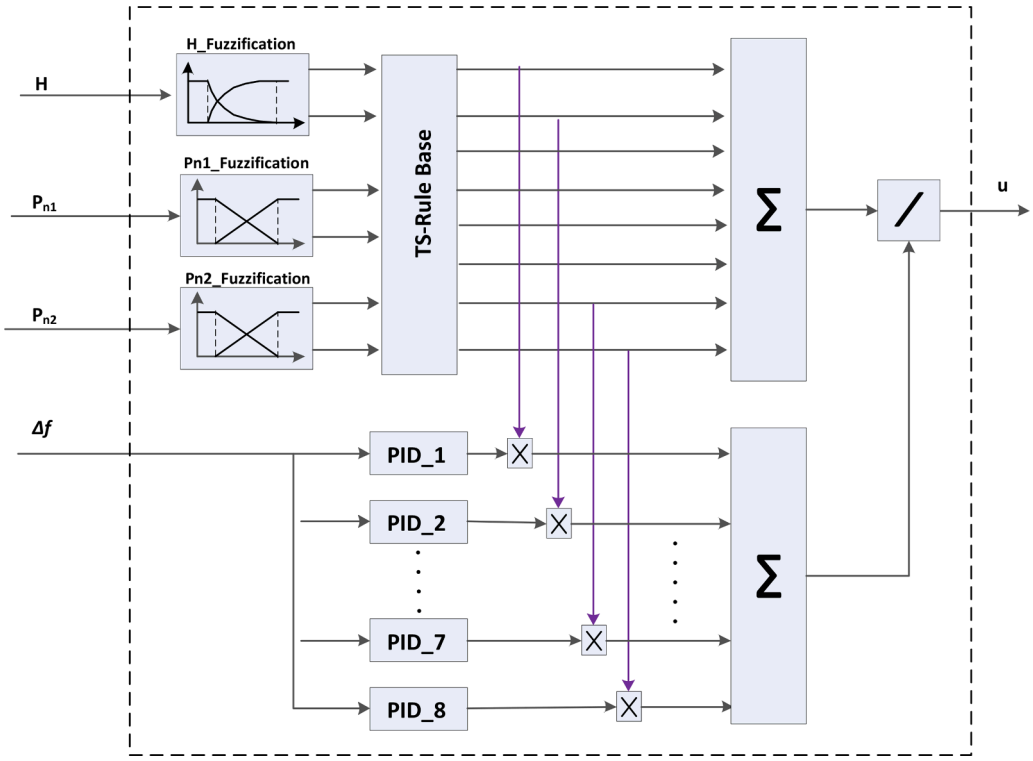


Fig. 5. Proposed Parallel distributed compensator PDC-PID using Takagi-Sugino.

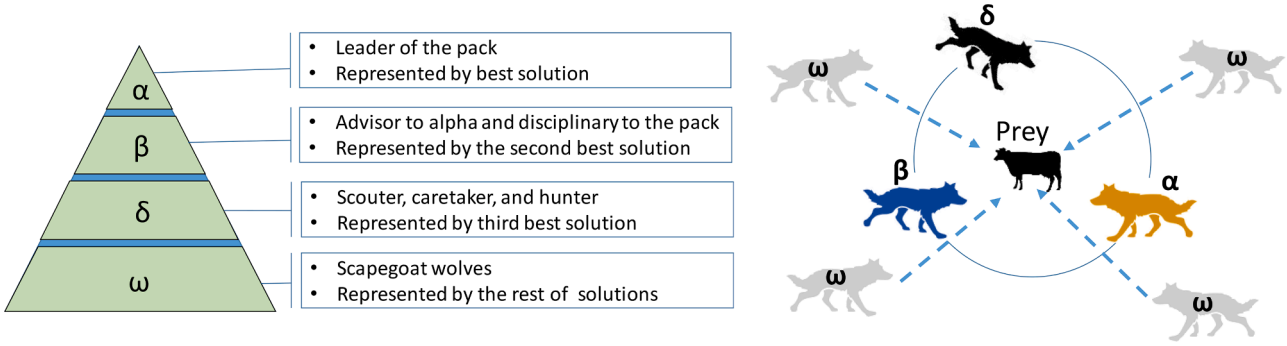


Fig. 6. Grey wolf pack organization, mapping, and hunting concept.

Table 4
GWO parameters and variables.

Number of search agents	20
Maximum number of iterations	20
Number of variables (problem dimension)	3
Lower boundaries	[0,0,0]
Upper boundaries	[200,20,100]

GWO, the third best value is denoted as delta wolf. The last category is the omega wolves which are all wolves that don't belong to the first three categories. The omega wolves follow the commands of the other categories. The rest solutions, from the fourth to the last solution, belong to this category. The exact location of the prey is not known, therefore the locations of the alpha, beta, and delta are used to estimate the location of the prey. After that all the pack moves toward this location as shown in Fig. 6. The parameters of the GWO technique are given in Table 4. The optimal PID controllers at the polytope vertices are given in Table 5

Table 5
Optimal PID controller gains at different vertices.

Gains	Operating point (Vertex) (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
K_p	85.727	171.2926	132.9505	108.1458	169.7781	200	200	106.9972
K_i	11.8253	2.7085	3.3129	2.1010	20	4.3221	5.9952	4.1072
K_d	30.6351	100	55.6911	80.9923	50.84	100	100	57.5891

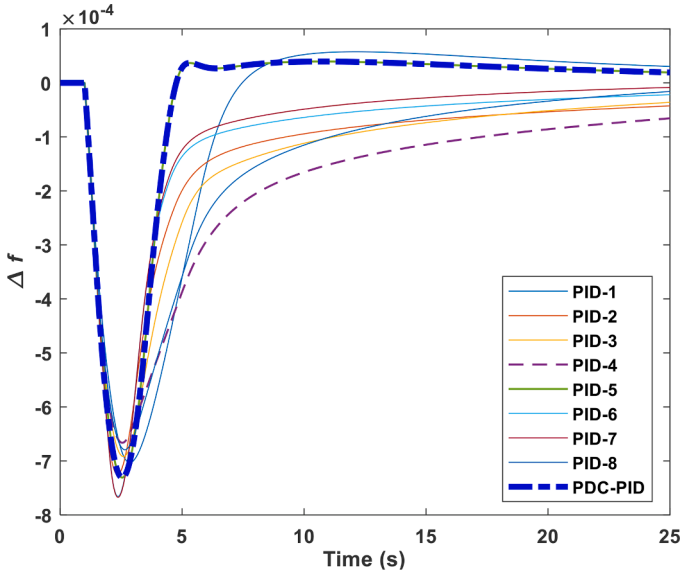


Fig. 7. Frequency response at point 5 (PID_5 & PDC are optimal).

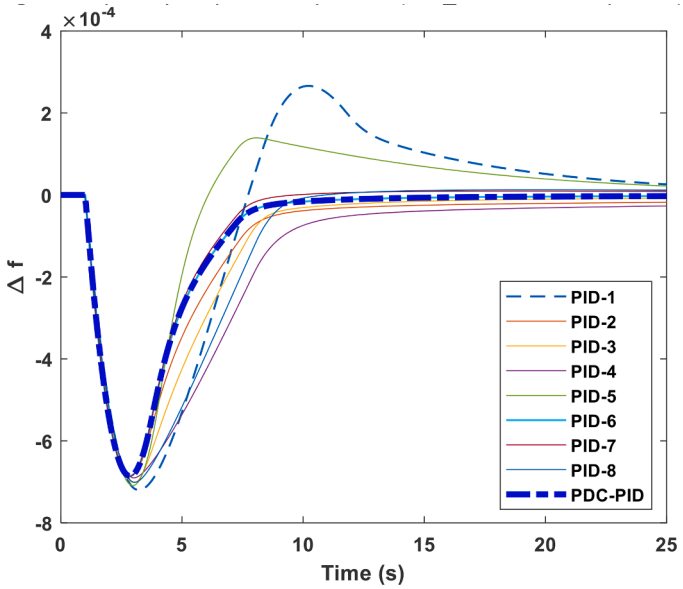


Fig. 9. Frequency response at point 6 (PID_6 & PDC are optimal).

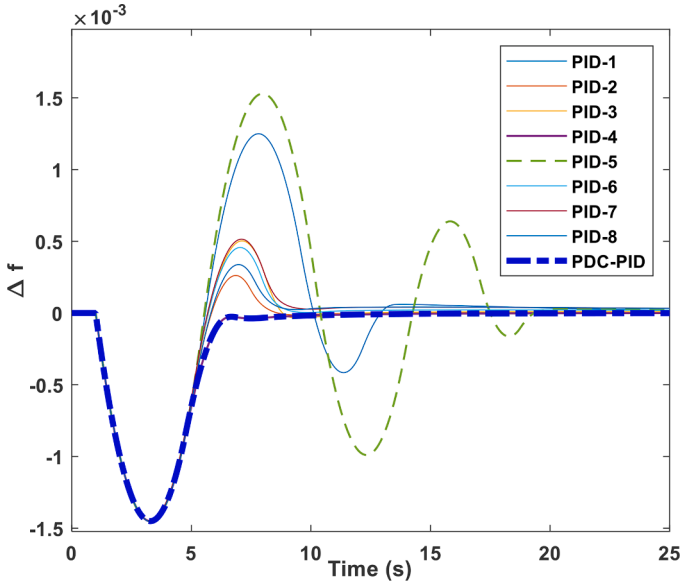


Fig. 8. Frequency response at point 4 (PID_4 & PDC are optimal).

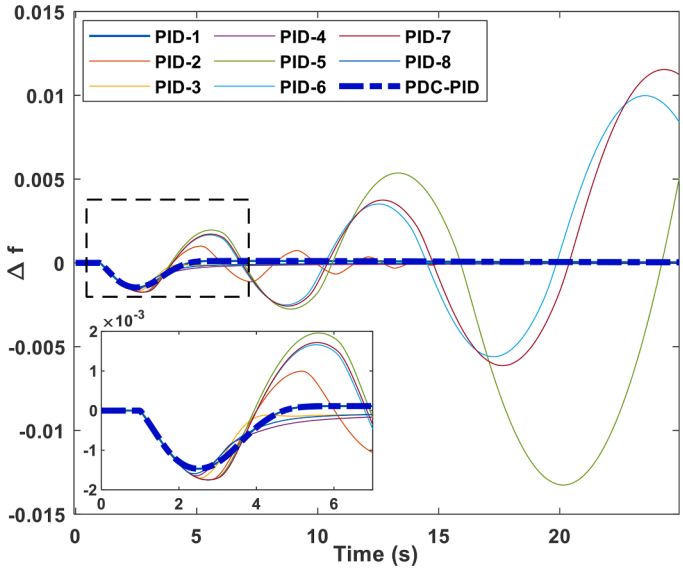


Fig. 10. Frequency response at point 1 (PID_1 & PDC are optimal).

4. Simulation results

Different simulation cases are performed to compare the response of the proposed PDC-PID with fixed PID controllers of different gains. The comparison is implemented at polytope vertices and other operating points inside the polytope representation.

4.1. Operation at polytope vertices

In this case, the control system is tested at eight operating points which represent the vertices of the system polytope, with a step change in load by 1% at 1 s. At operating point (5), the optimal response is obtained using optimal PID_5, while the worst response is achieved with controller PID_4. The PDC controller provides a similar response as PID_5 as shown in Fig. 7. On the other hand, at operating point (4), the controller PID_5 provides higher overshoot. However, the optimal response is achieved using optimal controller PID_4 and PDC-PID as shown in Fig. 8. A more remarkable achievement is observed if the system response is compared at point (6) and point (1). Since, at point (6), the optimal system response is achieved with PID_6, while PID_1 exhibits higher overshoot and longer settling time. in contrast at point (1), the optimal response is obtained using PID_1, while the PID_6 losses the system stability as well as PID-5, and PID_7. Therefore, with change in the system operating conditions, fixed gain controller may provide unsatisfactory response with higher overshoot, higher oscillation, longer settling time, or losing the system stability. The system response at the remaining vertices are shown in Figs. 9, 11, 12, 13 and 14. It is clear, at polytope vertices, the proposed PDC is capable of providing the optimal response similar to the optimal controller corresponding to the operating conditions.

The proposed PDC is able to achieve the minimum value of the proposed objective function at different operating point as well as the local optimal controllers as shown in Table 6. Also, the system with the suggested PDC has lower ITAE at different operating conditions is given in Table 7.

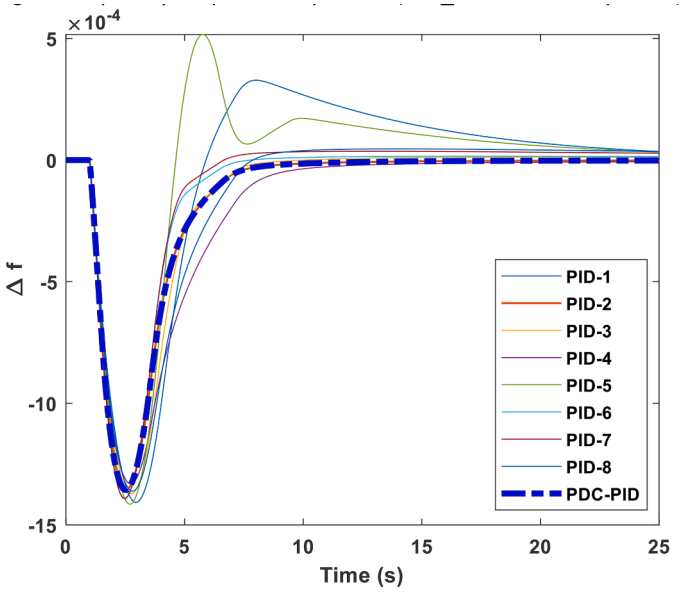


Fig. 11. Frequency response at point 2 (PID_2 & PDC are optimal).

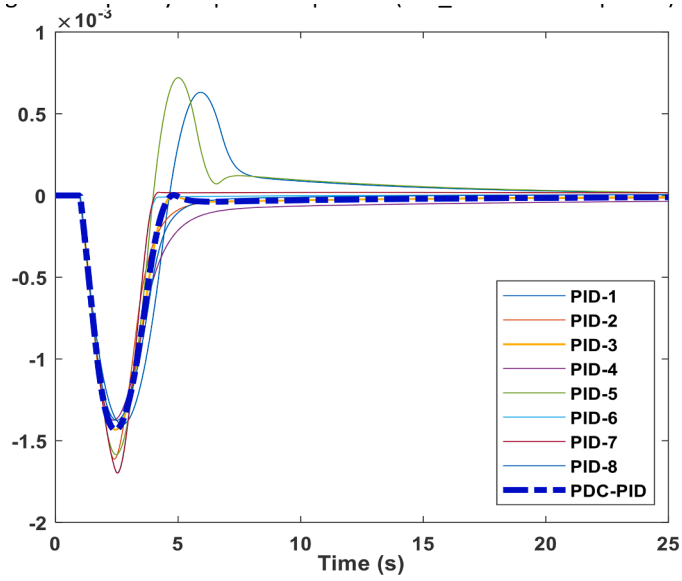


Fig. 12. Frequency response at point 3 (PID_3 & PDC are optimal).

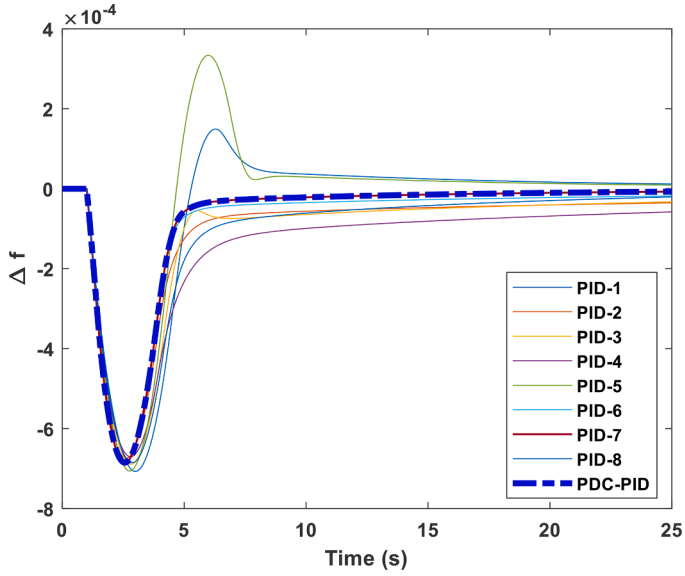


Fig. 13. Frequency response at point 7 (PID_7 & PDC are optimal).

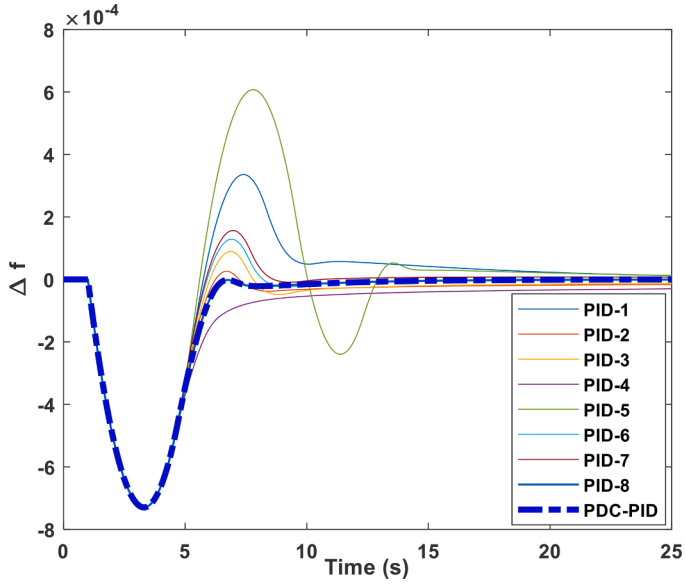


Fig. 14. Frequency esponse at point 8 (PID_8 & PDC are optimal).

4.2. *Operating point inside the polytope*

To check the ability of the designed parallel distributed compensator to provide a better response, the system is subjected to disturbance while the system is operating at the average point inside the polytope cube (P_{n1} = 0.225, P_{n2} = 0.42, *and* H = 6). The load is changing in steps as shown in Fig. 15. At load increase the power system frequency is reduced. While at load decrease the generation becomes greater than the demand consequently, the power system frequency increase as shown in Fig. 16. The proposed PDC controller succeeded to provide damped response with fast recovery time.

5. Robust PID controller design based on H_∞ using GWO-ILMI-PID

The closed control system of robust controller is shown in Fig. 17. Since, u, W, y, and z_∞, are the control input, disturbance input, the system output, and the robustness indicator variable (fictitious vector). To minimize the order of H_∞ controller, the controller is assumed to be PID controller which satisfies the requirements of robust control [14] which converts the robust PID control problem into static output feedback optimization problem solved by iterative linear matrix inequality ILMI as elaborated in this section.

The open loop uncontrolled system with disturbance is given by Eq. (67) and (68). Where $A = A_p$, $B_2 = B_p$ *and* $B_1 = E_p$, $C = C_p$ at the average system operating point. $x = x_p$ *and* $y = y_p$. After adding integrator state, the system state increases by one as shown in Eq. (69). The output of PID controller is obtained by combination of the system output (y_1), the integrator output (y_2), and derivative of the output (y_3) as shown in Eq. (70), where $F_1 = K_p$, $F_2 = K_i$, *and* $F_3 = K_d$ are the PID gains, y_s is the sensed or the controller feedback signals as shown by Eq. (71). The details for conversions of the PID control action into the form of Eq. (70) can be found in [14,40].

$$\dot{x} = Ax + B_1 W + B_2 u \tag{67}$$

Table 6
Objective function value operating at polytope vertices.

Controller	$OF = 10\|\Delta f(t)\|_{max} + 10\max(\Delta f(t)) + 0.1\int_0^t t\|\Delta f(t)\|\ dt$							
	point_1	point_2	point_3	point_4	point_5	point_6	point_7	point_8
PID_1	0.0212	0.0225	0.0242	0.0326	0.0128	0.0140	0.0113	0.0127
PID_2	0.0339	0.0148	0.0189	0.0198	0.0127	0.0115	0.0135	0.0125
PID_3	0.0228	0.0177	0.0169	0.0232	0.0135	0.0090	0.0112	0.0113
PID_4	0.0232	0.0155	0.0190	0.0158	0.0153	0.0129	0.0159	0.0151
PID_5	**unstable**	0.0234	0.0262	0.0403	0.0107	0.0113	0.0124	0.0164
PID_6	**unstable**	0.0182	0.0194	0.0250	0.0115	0.0080	0.0094	0.0098
PID_7	**unstable**	0.0200	0.0214	0.0270	0.0119	0.0096	0.0087	0.0108
PID_8	0.0242	0.0208	0.0179	0.0239	0.0153	0.0109	0.0108	0.0080
Proposed	**0.0212**	**0.0148**	**0.0169**	**0.0158**	**0.0107**	**0.0080**	**0.0087**	**0.0080**

Table 7
ITAE at different operating point of the system vertices.

Controller	$ITAE = \int_0^t t\|\Delta f(t)\|\ dt$							
	point_1	point_2	point_3	point_4	point_5	point_6	point_7	point_8
PID_1	0.0542	0.0514	0.0384	0.0551	0.0516	0.0415	0.0272	0.0206
PID_2	0.0641	0.0120	0.0276	0.0266	0.0541	0.0462	0.0677	0.0497
PID_3	0.0566	0.0389	0.0256	0.0361	0.0649	0.0192	0.0425	0.0315
PID_4	0.0662	0.0223	0.0525	0.0127	0.0855	0.0595	0.0920	0.0778
PID_5	**unstable**	0.0409	0.0313	0.1053	0.0302	0.0279	0.0202	0.0300
PID_6	**unstable**	0.0411	0.0238	0.0590	0.0383	0.0114	0.0252	0.0128
PID_7	**unstable**	0.0573	0.0426	0.0738	0.0417	0.0262	0.0186	0.0192
PID_8	0.0808	0.0678	0.0405	0.0598	0.0838	0.0376	0.0391	0.0072
Proposed	**0.0542**	**0.0120**	**0.0256**	**0.0127**	**0.0302**	**0.0114**	**0.0186**	**0.0072**

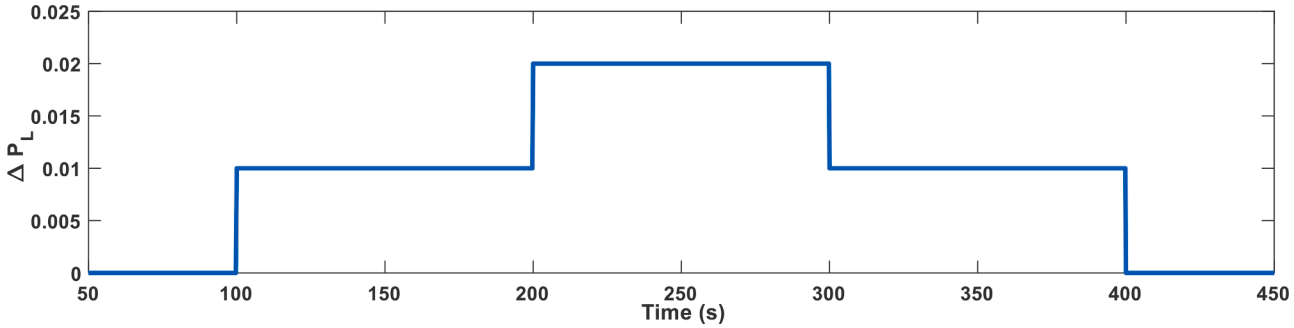


Fig. 15. Load disturbance with average system model.

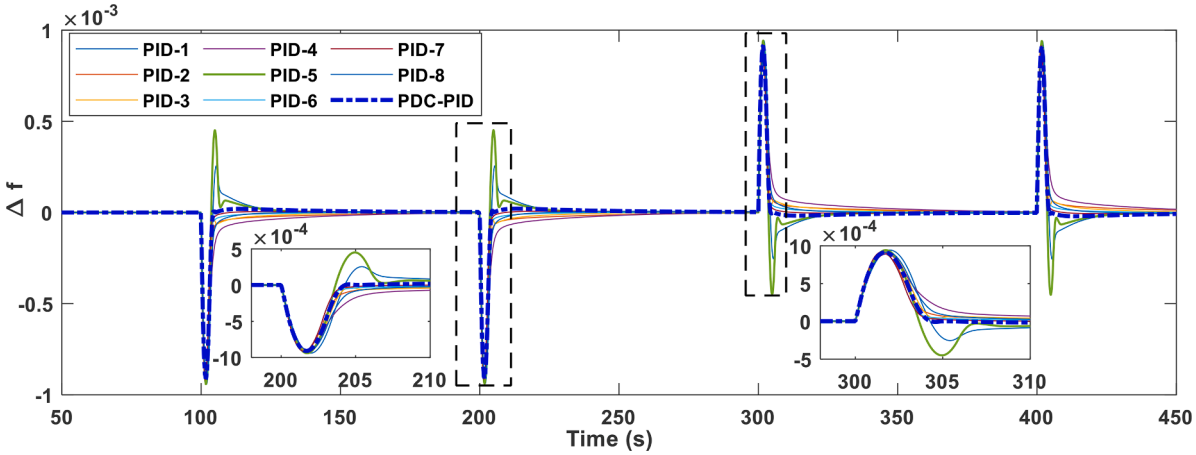


Fig. 16. Frequency response with average system model.

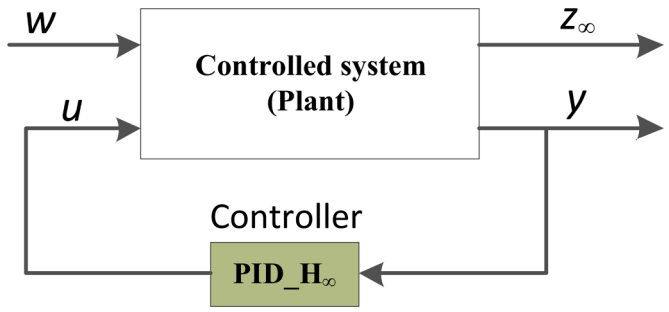


Fig. 17. Robust controller design block diagram [14].

$$y = Cx \tag{68}$$

$$z_1 = x \text{ and } z_2 = \int y dt \tag{69}$$

$$u = (I - F_3 C B_2)^{-1} [F_1 \; F_2 \; F_3] \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \overline{F} y_s \tag{70}$$

$$y_s = \begin{bmatrix} C & 0 \\ 0 & I \\ CA & 0 \end{bmatrix} z = \overline{C}_s z \tag{71}$$

From (67), (70) and (71), The closed loop system state equation is given by Eq. (72). For robust H_∞ controller, the objective is minimization the effect of exogenous inputs W on the performance indicator z_∞. The performance indicator can be represented as function of the system states and input as given by Eq. (73).

$$\dot{z} = \overline{A} z + \overline{B}_1 W + \overline{B}_2 \overline{F} \overline{C}_s z = (\overline{A} + \overline{B}_2 \overline{F} \overline{C}_s) z + \overline{B}_1 W \tag{72}$$

$$z_\infty = \overline{C}_r z + \overline{D}_r u = \overline{C}_r z + \overline{D}_r \overline{F} y_s = \overline{C}_r z + \overline{D}_r \overline{F} C_s z = (\overline{C}_r + \overline{D}_r \overline{F} \overline{C}_s) z \tag{73}$$

$$\overline{A} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix}, \; \overline{B}_1 = \begin{bmatrix} B_1 \\ 0 \end{bmatrix}, \text{ and } \overline{B}_2 = \begin{bmatrix} B_2 \\ 0 \end{bmatrix}$$
$$z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}, \text{ and } \overline{C}_s = \begin{bmatrix} C & 0 \\ 0 & I \\ CA & 0 \end{bmatrix} \tag{74}$$

The controller is suboptimal H_∞ controller if there is a positive definite matrix $P > 0$ satisfies the following LMIs (75) and (76).

$$\| \frac{Z_\infty(s)}{W(s)} \|_\infty < v \tag{75}$$

$$\begin{bmatrix} (\overline{A} + \overline{B}_2 \overline{F} \overline{C}_s)^T P + P(\overline{A} + \overline{B}_2 \overline{F} \overline{C}_s) & P\overline{B}_1 & (\overline{C}_r + \overline{D}_r \overline{F} \overline{C}_s)^T \\ \overline{B}_1^T P & -vI & 0 \\ (\overline{C}_r + \overline{D}_r \overline{F} \overline{C}_s) & 0 & -I \end{bmatrix} < 0 \tag{76}$$

The LMI is quadratic of the matrices P *and* F, the problem becomes nonconvex and can be solved using iterative linear matrix inequality (ILMI) algorithm which is originally suggested in [40] as follows:

Step 1: Choose the weighting matrix $Q_0 > 0$ and solve the following algebraic Ricatti Eq. (77) for P

$$\overline{A}^T P + P\overline{A} - P\overline{B}_2 \overline{B}_2^T P + Q_0 = 0 \tag{77}$$

Step2: Set $i = 1$ *and* $X_i = P$ (P *is obtained solution of Ricatti*) and Solve the following optimization problem shown by Eq. (78) for P_i, F, *and* α_i Subject to LMI (79).

$$\min \alpha_i \tag{78}$$

$$\begin{bmatrix} \Sigma - \alpha_i P_i & P_i \overline{B}_1 & (\overline{C}_r + \overline{D}_r \overline{F} \overline{C}_s)^T & \left(\overline{B}_2^T P_i + \overline{F} \overline{C}_s\right)^T \\ \overline{B}_1^T P_i & -vI & 0 & 0 \\ (\overline{C}_r + \overline{D}_r \overline{F} \overline{C}_s) & 0 & -I & 0 \\ \left(\overline{B}_2^T P_i + \overline{F} \overline{C}_s\right) & * & 0 & -I \end{bmatrix} < 0 \tag{79}$$

$$\Sigma = \overline{A}^T P_i + P_i \overline{A} - X_i \overline{B}_2 \overline{B}_2^T P_i - P_i \overline{B}_2 \overline{B}_2^T X_i + X_i \overline{B}_2 \overline{B}_2^T X_i \tag{80}$$

Step 3: If $\alpha_i^* > 0$ go to step 4, while If $\alpha_i^* < 0$, where α_i^* is the optimal value of step 2, hence the obtained matrix F is the solution of the H_∞ suboptimal controller and stop ILMI procedure. The PID controller gains are determined using Eq. (81) to (84).

$$F_3 = \overline{F}_3 (I + C B_2 \overline{F}_3)^{-1} \tag{81}$$

$$F_2 = (I - F_3 C B_2) \overline{F}_2 \tag{82}$$

$$F_1 = (I - F_3 C B_2) \overline{F}_1 \tag{83}$$

$$K_p = F_1, K_i = F_2, \text{ and } K_d = F_3 \tag{84}$$

Note: the necessary condition for satisfying this methodology is the matrix $(I + CB_2 \overline{F}_3)$ to be invertible. If $CB_2 = 0$, this condition is satisfied.

Step 4: Solve the following optimization problem given by (85) (determine $P_i > 0$), subject to (86) or (79) at $\alpha_i = \alpha_i^*$

$$\min tr(P_i) \tag{85}$$

$$\begin{bmatrix} \Sigma - \alpha_i P_i & P_i \overline{B}_1 & (\overline{C}_r + \overline{D}_r \overline{F} \overline{C}_s)^T & \left(\overline{B}_2^T P_i + \overline{F} \overline{C}_s\right)^T \\ \overline{B}_1^T P_i & -vI & 0 & 0 \\ (\overline{C}_r + \overline{D}_r \overline{F} \overline{C}_s) & 0 & -I & 0 \\ \left(\overline{B}_2^T P_i + \overline{F} \overline{C}_s\right) & * & 0 & -I \end{bmatrix} < 0 \tag{86}$$

$$\Sigma = \overline{A}^T P_i + P_i \overline{A} - X_i \overline{B}_2 \overline{B}_2^T P_i - P_i \overline{B}_2 \overline{B}_2^T X_i + X_i \overline{B}_2 \overline{B}_2^T X_i \tag{87}$$

Step 5: If $\| X_i B_1 - P_i^* B_1 \|_2 < \varepsilon$, where ε is a predefined tolerance, go to step 6. Otherwise $i = i + 1$, $X_i = P_i^*$ and go to step 2.

Step 6: It cannot be decided by this algorithm whether SOFH∞ problem is solvable and stop ILMI procedure.

The ILMI procedure has two tuning parameters (v *and* Q_0) which effect on the system performance. Therefore, the optimal values are $Q_0 = 16.7789$ and $v = 39.4157$ which are determined using GWO algorithm to optimize the system ITAE, nadir and overshoot as shown by Eq. (66). $K_p = 94.3547$, $K_i = 16.3258$, and $K_d = 221.503$. The comparisons between the system dynamics with the proposed PDC-PID and the optimal PID_H∞ at step load disturbance are shown in Figs. 18 to 26.

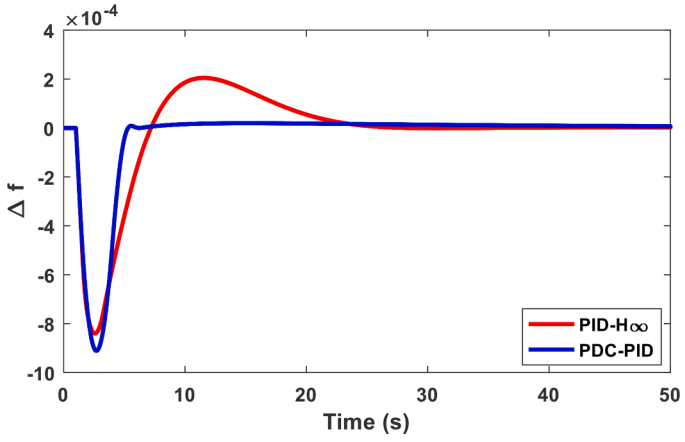


Fig. 18. Frequency response at average operating point.

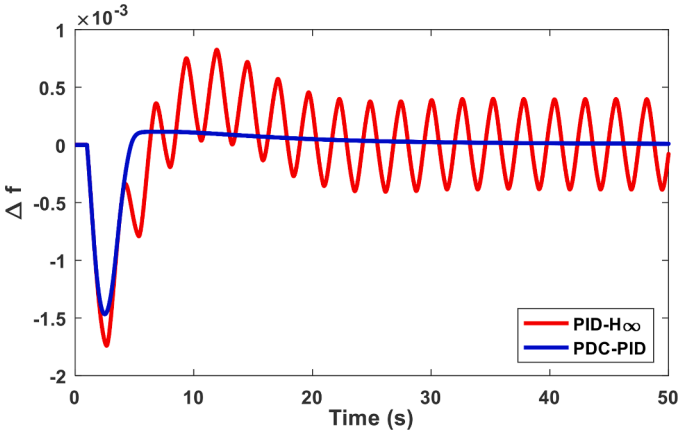


Fig. 19. Frequency response at operating point 1.

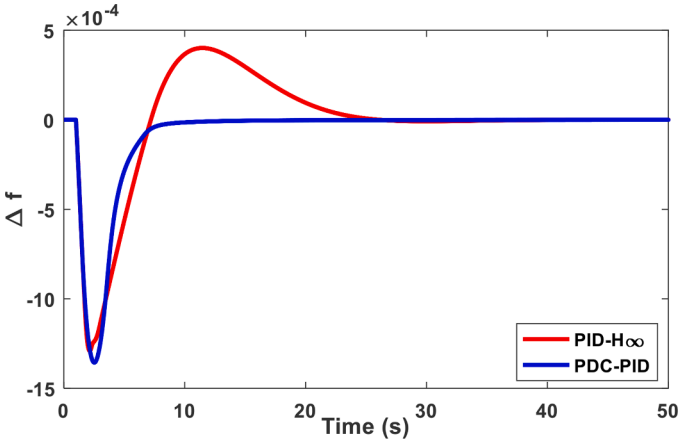


Fig. 20. Frequency response at operating point 2.

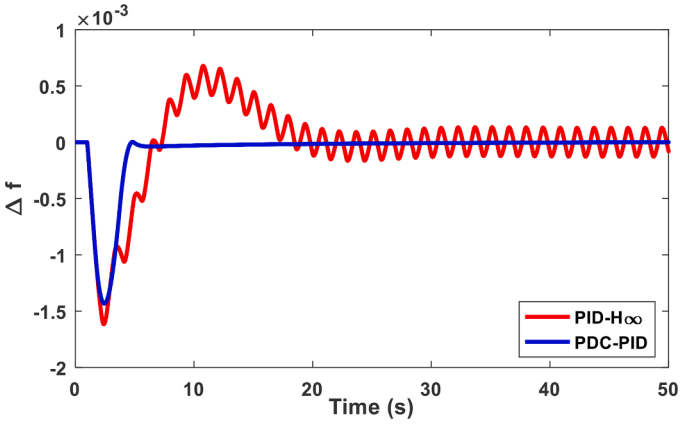


Fig. 21. Frequency response at operating point 3.

At the average operating point, the PID_H_∞ provides less frequency nadir compared to the PDC-PID while the system response becomes slower with higher overshoot. When the operating point changes to point (1) or point (3), due to the presence of system nonlinearities, the PID_H_∞ has an oscillating response as shown in Fig. 19 and 21. Even the PID_H_∞ is stable at certain operating points, the system suffers from higher overshot and longer settling time. Fig. 27 shows the system response at average operating point with parameter variation by increasing the time constant ($T1$, $T2$, *and* $T3$) by 50% and reduction in the permissible generation rate ($GRC1$, $GRC2$, *and* $GRC3$) by 25% . It is

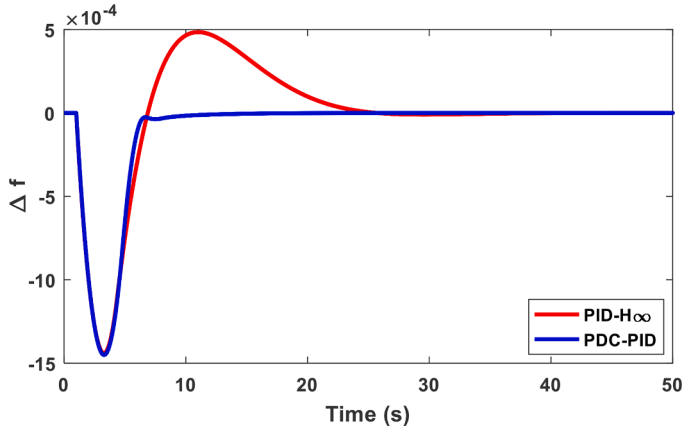


Fig. 22. Frequency response at operating point 4.

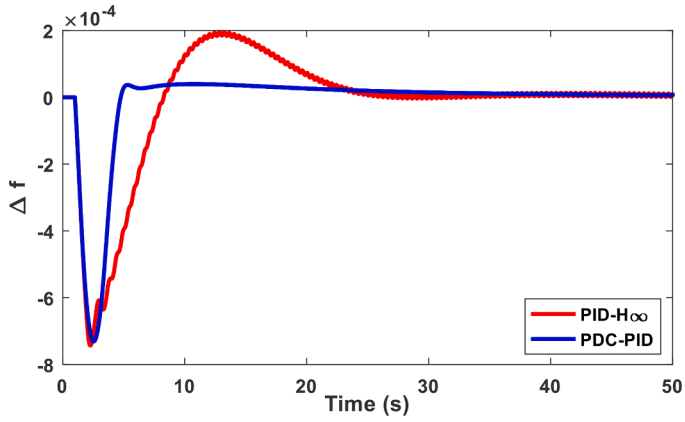


Fig. 23. Frequency response at operating point 5.

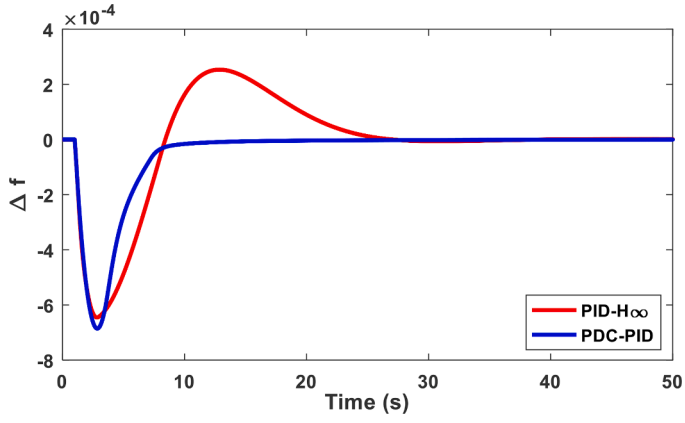


Fig. 24. Frequency response at operating point 6.

clear, the proposed PDC is able to provide faster response and better disturbance rejection compared to the robust PID_H∞

6. Stability analysis of the closed-loop system

In order to provide graphical evidence that the proposed control strategy of the proposed controller section 3) is able to maintain the system stability over a wide range of parameter variations, the closed-loop system poles are located in the complex plan under PDC-PID is examined at a large number of operating points and located in the complex S-plan. To determine these poles, the closed system state space

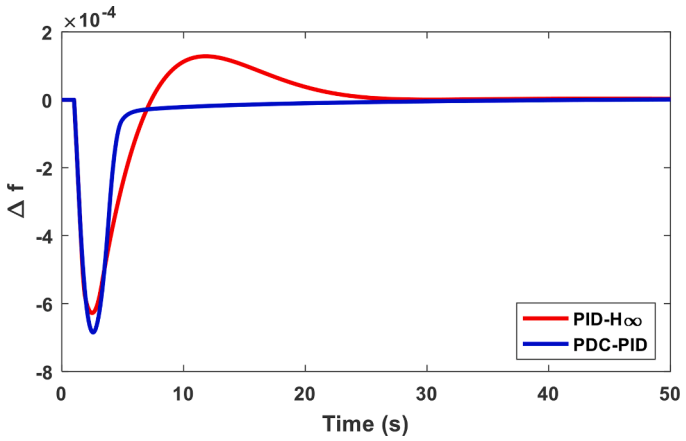


Fig. 25. Frequency response at operating point 7.

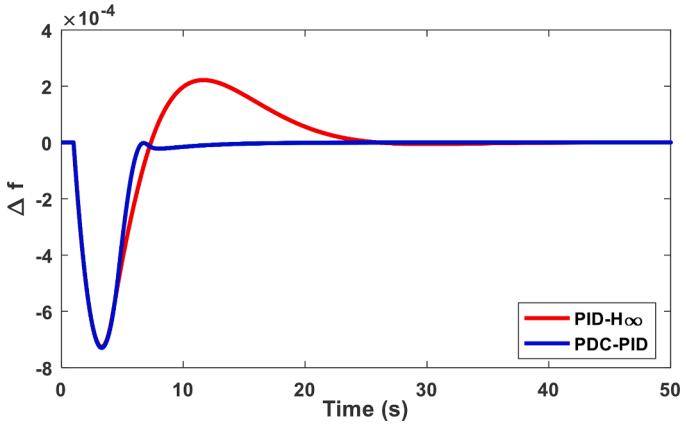


Fig. 26. Frequency response at operating point 8.

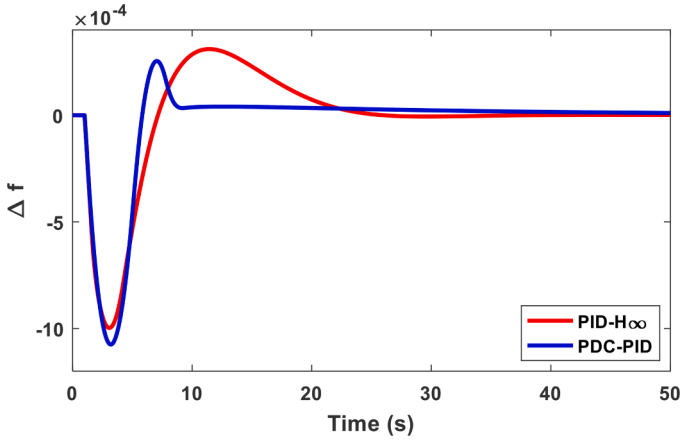


Fig. 27. Frequency response at average operating point with parameter variation.

model shall be determined considering the proposed PDC effect. To make this task not complicated, the fact that the PDC is considered a PID controller with variable gains. The output of PDC is the combination of the output of the eight optimal PID controllers, as shown by Eq. (88) and ((89). Eq. (89) can be re-arranged as shown in Eq. (90). Eq. (90) comprises three terms, the first term is proportional to the deviation of the frequency, the second term is proportional to the integral of frequency deviation, and the third term is proportional to the derivative of the frequency. Hence, the proposed PDC is an PID controller with variable gains. The equivalent gains (K_{pe}, K_{ie}, *and* K_{de}) can be determined by comparing Eq. (90) with the standard form of the PID controller of Eq. (91). The resultant equivalent gains are given in Eq. (92) -(94). It should be noted that these gains are varying according to the optimal controllers' gains at vertices and the system operating point.

$$u(t) = \sum_{j=1}^{R} W_j u_j(t) \tag{88}$$

$$u(t) = \sum_{j=1}^{R} W_j \left[K_{pj}\Delta f(t) + K_{ij}\int \Delta f(t)dt + K_{dj}\frac{d\Delta f(t)}{dt} \right] \tag{89}$$

$$u(t) = \left(\sum_{j=1}^{R} W_j K_{pj} \right) \Delta f(t) + \left(\sum_{j=1}^{R} W_j K_{ij} \right) \int \Delta f(t)dt + \left(\sum_{j=1}^{R} W_j K_{dj} \right) \frac{d\Delta f(t)}{dt} \tag{90}$$

$$u(t) = K_{pe}\Delta f(t) + K_{ie}\int \Delta f(t)dt + K_{de}\frac{d\Delta f(t)}{dt} \tag{91}$$

$$K_{pe} = \sum_{j=1}^{R} W_j K_{pj} \tag{92}$$

$$K_{ie} = \sum_{j=1}^{R} W_j K_{ij} \tag{93}$$

$$K_{de} = \sum_{j=1}^{R} W_j K_{dj} \tag{94}$$

The derivative of system frequency can be obtained from the swing equation in time domain as given by Eq. (95). Substitute from (95) into (90) regarding the derivative of the system frequency, the control signal is given in Eq. (96). The integral part of the controller is considered as a new state in the closed loop system as shown by Eq. (97). The corresponding derivative of the new state is given by Eq. (98). The compact form of the control input is given by Eq. (99).

$$\frac{d\Delta f(t)}{dt} = -\frac{D}{2H}\Delta f(t) + \frac{1}{2H}\Delta P_{m1}(t) + \frac{1}{2H}\Delta P_{m2}(t) + \frac{1}{2H}\Delta P_{m3}(t) - \frac{1}{2H}P_L(t) \tag{95}$$

$$u(t) = K_{pe}\Delta f(t) + K_{ie}\int \Delta f(t)dt + K_{de}\left(-\frac{D}{2H}\Delta f(t) + \frac{1}{2H}\Delta P_{m1}(t) + \frac{1}{2H}\Delta P_{m2}(t) + \frac{1}{2H}\Delta P_{m3}(t) \right) - \frac{K_{de}}{2H}P_L(t) \tag{96}$$

$$x_c(t) = \int \Delta f(t)dt \tag{97}$$

$$\dot{x}_c(t) = \Delta f(t) = x_1(t) = Z_c x_p(t) \tag{98}$$

$$Z_c = [1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]u(t) = Z_1 x_p(t) + K_{ie}x_c(t) + Z_3 W(t) \tag{99}$$

$$Z_1 = \left[\left(K_{pe} - \frac{K_{de}D}{2H} \right) \quad \frac{K_{de}}{2H} \quad \frac{K_{de}}{2H} \quad 0 \quad \frac{K_{de}}{2H} \quad 0 \right]$$
$$Z_3 = -\frac{K_{de}}{2H} \tag{100}$$

Substitute from (99) into (13) regarding the control input, the plant state equation is given in Eq. (101). Eqs. (101) and (98) describe the complete state equation of the closed-loop system as given in Eq. (102). The closed-loop poles of the system are the eigenvalues of the closed-loop system matrix A_{cl} of Eq. (103).

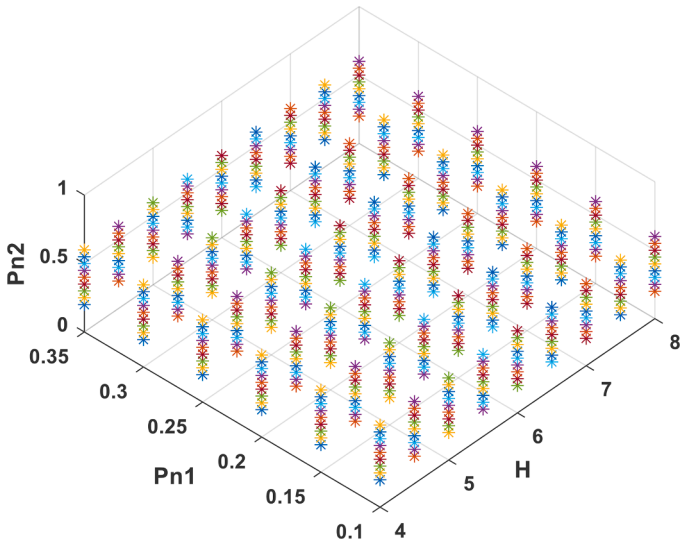


Fig. 28. Assumed different 486 operating points.

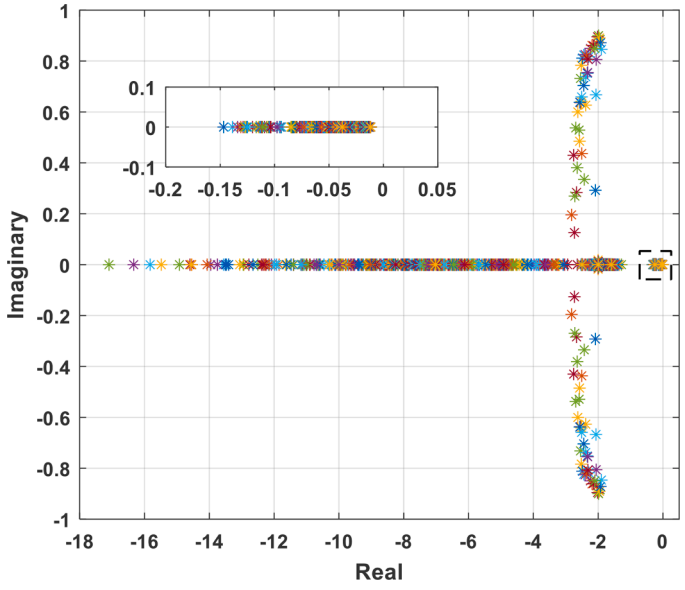


Fig. 29. Closed loop poles trajectory for 486 different operating points with PDC-PID.

$$\dot{x}_P(t) = (A_P + B_P Z_1)x_P(t) + B_P K_{ie} x_c(t) + \left(E_p + B_P Z_3\right)\Delta P_L(t) \tag{101}$$

$$\begin{bmatrix} \dot{x}_P(t) \\ \dot{x}_C(t) \end{bmatrix} = \begin{bmatrix} A_P + B_P Z_1 & B_P K_{ie} \\ Z_c & 0 \end{bmatrix} \begin{bmatrix} x_P(t) \\ x_c(t) \end{bmatrix} + \begin{bmatrix} E_p + B_P Z_3 \\ 0 \end{bmatrix} W(t) \tag{102}$$

$$A_{cl} = \begin{bmatrix} A_P + B_P Z_1 & B_P K_{ie} \\ Z_c & 0 \end{bmatrix} \tag{103}$$

To check the ability of the designed parallel distributed compensator to maintain the system stable at different operating conditions inside the cubical polytope representation, the system closed loop poles are obtained at 486 different operating conditions for variation of inertia, nominal power of non-reheat, and nominal power of reheat by 0.5, 0.05 and 0.05 respectively as shown in Fig. 28. The closed-loop poles of the different operating conditions are illustrated in Fig. 10. From Fig. 29, the major part of closed-loop poles lay on the real axis which produces higher damping and less oscillation.

The advantages and drawbacks of the proposed PDC-PID controller can be summarized as follow:

- Advantages:

  The proposed controller

  - Provides optimal response at the polytope system vertices
  - Guarantees the system stability over the bounded space of the parameter variation
  - Exhibits better response compared to optimal fixed gains controller.
  - Enhances the system response compared to robust fixed gain PID controller.
  - Implantation is straightforward and physical meaning is clear, because It is similar to the conventional PID with adaptive gains as proved in the paper.
- Drawbacks:
  - The proposed controller depends on the system variables which are the running capacity and running inertia of the system.

7. Conclusion

The article presented load frequency control (LFC) of the Egyptian power system as a real multi-generation integration that has varying generation capacity according to the operating conditions. The proposed control is parallel distributed compensator (PDC) based on PID and Takagi-Sugino fuzzy system, which comprises multiple local optimal controllers and uses fuzzy blending to produce the final output of the controller. The simulation results show that the design of a single optimal PID controller or single robust PID controller at a particular operating condition may provide unsatisfactory performance at change in the working environments. However, the proposed controller succeeded to provide optimal operation at a set of operating conditions called the system vertices.

The local controllers are optimized concerning system nadir, oscillation, and integral time absolute error using the grey wolf optimizer technique while the stability conditions are included in terms of problem constraints. Therefore, the proposed controller provides less ITAE with fast and damped response compared to fixed gain optimal controllers and fixed gain robust PID controller. The obtained closed-loop system over a wide range of parameter variation has a dominant pole on the real axis, which emphasizes the system damping as intended by using the proposed objective function.

The proposed PDC controller succeeded to exhibit better disturbance rejection performance compared to fixed gains robust PID controllers for exogenous inputs disturbance (load disturbance), and endogenous disturbances due to system's parameters uncertainties and nonlinearities.

The paper developed sufficient conditions to maintain the system's stability over the entire varying parameters' space using polytope system representation. The stability conditions are expressed using a set of linear matrix inequalities. The number of LMIs is increasing dramatically with the number of polytope system vertices. Hence, the paper utilized tensor product transformation technique to produce an accurate system with a minimum number of vertices. The resultant tensor product weighting functions are utilized as the membership functions of the fuzzy blender system to reduce the system conservatism for stability. consequently, the number of LMIs is reduced from 512 to 120 with more relaxation.

Eventually, the system's stability over a wide range of parameters variation has been verified using closed-loop poles locations in the S-plane. The closed loop system poles are obtained by developing a PID controller equivalent to the PDC-PID controller. The equivalent PID is a nonlinear controller, since the controller gains (K_{pe}, K_{ie}, *and* K_{de}) are varying according to the operating condition of the system. The closed loop poles are located in the left-hand side of the S-plane, which indicates the stability of the proposed control system.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix

List of abbreviations and symbols

LFC	Load Frequency Control
ITAE	Integral Time Absolute Error
PDC	Parallel Distributed Compensator
TS	Takagi-Sugino
LMI	Linear Matrix Inequality
ILMI	Iterative Linear Matrix Inequality
GWO	Grey Wolf Optimizer
GRC	Generation Rate Constraints
LPV	Linear Parameter Varying
OF	Objective function
$x_P(t)$	Open loop system states vector
$u(t)$	Control input vector
$W(t)$	Input disturbance vector
$y(t)$	System output
$p(t)$	Variable parameters
A or A_P or $A_P(p(t))$	Open loop state matrix
B_2or B_P or $B_P(p(t))$	Open loop input matrix
B_1or E_p or $E_p(p(t))$	Open loop disturbance matrix
C	Output matrix
D	Transfer matrix
R	Number of polytope vertices
r	Vertices number
$W_r(p(t))$	Weight of vertex r
A_r	System matrix at vertex r
B_r	System input matrix at vertex r
$W_{1,i}(p_1(t))$	grade of variables $p_{n1}(t)$ in membership i
$W_{2,j}(p_2(t))$	grade of variables $p_{n2}(t)$, in membership j
$W_{3,k}(p_3(t))$	grade of variables $H(t)$, in membership k
$A_{i,j,k}$	System matrix corresponding to weighting function $(i, j,$ and $k)$
$B_{i,j,k}$	System matrix corresponding to weighting function $(i, j,$ and $k)$
ε	Error between the polytope model and nonlinear model
K_{pj}	Proportional gain at rule j
K_{ij}	Integral gain at rule j
K_{dj}	Derivative gain at rule j
$x_c(t)$	Controller state
$x(t)$	Closed loop state vector
$A(p(t))$	Open System matrix with controller state
$B(p(t))$	Open System input matrix with controller state
$u_j(t)$	Controller output of rule j
$u_{1j}(t)$	First part of controller output of rule j
$u_{2j}(t)$	second part of controller output of rule j
F_j	Feedback gains at rule j
h_i	Weighting of vertex i
W_j	Weighting of vertex j
M_k	Weighting of vertex k
S	Lyapunov matrix
k_1, k_2, and k_3	Objective function weighting factors
K_{pe}	Equivalent proportional gain
K_{ie}	Equivalent integral gain
K_{de}	Equivalent derivative gain
Z_1, Z_3, and Z_c	Intermediate manipulation vectors

References

[1] Fayiz Alfaverh, Mouloud Denaï, Yichuang Sun, Optimal vehicle-to-grid control for supplementary frequency regulation using deep reinforcement learning, Electr. Power Syst. Res. 214 (2023), 108949.

[2] Ana Fernández-Guillamón, Eduard Muljadi, Angel Molina-García, Frequency control studies: a review of power system, conventional and renewable generation unit modeling, Electr. Power Syst. Res. 211 (2022), 108191.

[3] Yogesh V. Hote, Shivam Jain, PID controller design for load frequency control: past, Present and future challenges, IFAC-PapersOnLine 51 (4) (2018) 604–609.

[4] Hassan Haes Alhelou, Mohamad-Esmail Hamedani-Golshan, Reza Zamani, Ehsan Heydarian-Forushani, Pierluigi Siano, Challenges and opportunities of load frequency control in conventional, modern and future smart power systems: a comprehensive review, Energies 11 (10) (2018) 2497.

[5] Musa Khan, Haishun Sun, Yingmeng Xiang, Di Shi, Electric vehicles participation in load frequency control based on mixed H2/H∞, Int. J. Electr. Power Energy Syst. 125 (2021), 106420.

[6] Yanhe Xu, Chaoshun Li, Zanbin Wang, Nan Zhang, Bing Peng, Load frequency control of a novel renewable energy integrated micro-grid containing pumped hydropower energy storage, IEEE Access 6 (2018) 29067–29077.

[7] Neelesh Kumar Gupta, Manoj Kumar Kar, Arun Kumar Singh, Design of a 2-DOF-PID controller using an improved sine–cosine algorithm for load frequency control of a three-area system with nonlinearities, Protect. Control Modern Power Syst. 7 (1) (2022) 1–18.

[8] Dalia Yousri, Thaniaknti Sudhakar Babu, Ahmed Fathy, Recent methodology based Harris Hawks optimizer for designing load frequency control incorporated in multi-interconnected renewable energy plants, Sustainable Energy, Grids Networks 22 (2020), 100352.

[9] Kangdi Lu, Wuneng Zhou, Guoqiang Zeng, Yiyuan Zheng, Constrained population extremal optimization-based robust load frequency control of multi-area interconnected power system, Int. J. Electr. Power Energy Syst. 105 (2019) 249–271.

[10] Gaber Magdy, G. Shabib, Adel A. Elbaset, Yasunori Mitani, Optimized coordinated control of LFC and SMES to enhance frequency stability of a real multi-source power system considering high renewable energy penetration, Protect. Control Mod. Power Syst. 3 (1) (2018) 1–15.

[11] Reza Alayi, Farhad Zishan, Seyed Reza Seyednouri, Ravinder Kumar, Mohammad Hossein Ahmadi, Mohsen Sharifpur, Optimal load frequency control of island microgrids via a PID controller in the presence of wind turbine and PV, Sustainability 13 (19) (2021) 10728.

[12] Marko Č. Bošković, Tomislav B. Šekara, Milan R. Rapaić, Novel tuning rules for PIDC and PID load frequency controllers considering robustness and sensitivity to measurement noise, Int. J. Electr. Power Energy Syst. 114 (2020), 105416.

[13] Kang-Di Lu, Zheng-Guang Wu, Resilient Event-Triggered Load Frequency Control for Cyber-Physical Power Systems Under DoS Attacks, IEEE Trans. Power Syst. (2022), https://doi.org/10.1109/TPWRS.2022.3229667.

[14] Shashi Kant Pandey, Nand Kishor, Soumya R. Mohanty, Frequency regulation in hybrid power system using iterative proportional-integral-derivative H∞ controller, Electr. Power Components Syst. 42 (2) (2014) 132–148.

[15] Hany M. Hasanien, Attia A. El-Fergany, Salp swarm algorithm-based optimal load frequency control of hybrid renewable power systems with communication delay and excitation cross-coupling effect, Electric Power Syst. Res. 176 (2019), 105938.

[16] Ahmed Fathy, Abdullah G. Alharbi, Recent approach based movable damped wave algorithm for designing fractional-order PID load frequency control installed in multi-interconnected plants with renewable energy, IEEE Access 9 (2021) 71072–71089.

[17] Dipayan Guha, Provas Kumar Roy, Subrata Banerjee, Adaptive fractional-order sliding-mode disturbance observer-based robust theoretical frequency controller applied to hybrid wind–diesel power system, ISA Trans. (2022).

[18] Vivek Patel, Dipayan Guha, Shubhi Purwar, Neural network aided fractional-order sliding mode controller for frequency regulation of nonlinear power systems, Comput. Electr. Eng. 96 (2021), 107534.

[19] Gomaa Haroun Ah, Yin-Ya Li, Ant lion optimized hybrid intelligent PID-based sliding mode controller for frequency regulation of interconnected multi-area power systems, Trans. Inst. Meas. Control 42 (9) (2020) 1594–1617.

[20] Mokhtar Shouran, Fatih Anayi, Michael Packianather, Monier Habil, Load frequency control based on the bees algorithm for the great britain power system, Designs 5 (3) (2021) 50.

[21] Neelamsetti Kirn Kumar, Rahul Sanmugam Gopi, Ramya Kuppusamy, Srete Nikolovski, Yuvaraja Teekaraman, Indragandhi Vairavasundaram, Siripireddy Venkateswarulu, Fuzzy Logic-Based Load Frequency Control in an Island Hybrid Power System Model Using Artificial Bee Colony Optimization, Energies 15 (6) (2022) 2199.

[22] Meysam. Gheisarnejad, An effective hybrid harmony search and cuckoo optimization algorithm based fuzzy PID controller for load frequency control, Appl. Soft Comput. 65 (2018) 121–138.

[23] Meysam Gheisarnejad, Mohammad Hassan Khooban, Secondary load frequency control for multi-microgrids: HiL real-time simulation, Soft Comput. 23 (14) (2019) 5785–5798.

[24] Mohammad Hassan Khooban, Taher Niknam, Frede Blaabjerg, Tomislav Dragičević, A new load frequency control strategy for micro-grids with considering electrical vehicles, Electric Power Syst. Res. 143 (2017) 585–598.

[25] Emre. Çelik, Design of new fractional order PI–fractional order PD cascade controller through dragonfly search algorithm for advanced load frequency control of power systems, Soft Comput. 25 (2) (2021) 1193–1217.

[26] Komboigo Charles, Naomitsu Urasaki, Tomonobu Senjyu, Mohammed Elsayed Lotfy, Lei Liu, Robust load frequency control schemes in power system using optimized PID and model predictive controllers, Energies 11 (11) (2018) 3070.

[27] Jitendra Sharma, Yogesh V. Hote, Rajendra Prasad, Robust PID load frequency controller design with specific gain and phase margin for multi-area power systems, IFAC-PapersOnLine 51 (4) (2018) 627–632.

[28] M.A. Ghany, Mohamed A. Shamseldin, Parallel distribution compensation PID based on Takagi-Sugeno fuzzy model applied on Egyptian load frequency control, Int. J. Electr. Comput. Eng. 10 (5) (2020) 5274.

[29] Yezheng Wang, Zidong Wang, Lei Zou, Hongli Dong, Multiloop decentralized H∞ fuzzy PID-like control for discrete time-delayed fuzzy systems under dynamical event-triggered schemes, IEEE Trans. Cybern. (2020).

[30] Mengmeng Dai, Jinxiang Wang, Nan Chen, Guodong Yin, Fuzzy steering assistance control for path following of the steer-by-wire vehicle considering characteristics of human driver. 2018 IEEE Intelligent Vehicles Symposium (IV), IEEE, 2018, pp. 892–897.

[31] Amine Dehak, Anh-Tu Nguyen, Antoine Dequidt, Laurent Vermeiren, Michel Dambrine, Reduced-Complexity Affine Representation for Takagi-Sugeno Fuzzy Systems, IFAC-PapersOnLine 53 (2) (2020) 8031–8036.

[32] Z. Petres, P. Baranyi, P. Korondi and H. Hashimoto, "Trajectory Tracking by TP Model Transformation: case Study of a Benchmark Problem," in IEEE Transactions on Industrial Electronics, vol. 54, no. 3, pp. 1654–1663, June 2007, doi: 10.1109/TIE.2007.89469.

[33] Elena-Lorena Hedrea, Radu-Emil Precup, Raul-Cristian Roman, Emil M. Petriu, Tensor product-based model transformation approach to tower crane systems modeling, Asian J. Control 23 (3) (2021) 1313–1323.

[34] Yihang Zuo, Bojun Liu, Kunming Shao, Symmetrical indoor visible light layout optimized by a modified grey wolf algorithm, Appl. Opt. 61 (20) (2022) 6016–6022.

[35] Seyedali Mirjalili, Seyed Mohammad Mirjalili, Andrew Lewis, Grey wolf optimizer, Adv. Eng. Software 69 (2014) 46–61.

[36] Abdullah Altay, Cem Şahin, Ires Iskender, Doğan Gezer, Caner Çakır, A compensator design for the aged hydroelectric power plant speed governors, Electric Power Syst. Res. 133 (2016) 257–268.

[37] Magdi A. Mosa, Ahmed A. Ali, Helmy M.El Zoghpy, Adel A.El Samahy, A.M. Abdel Ghany, Robust Control of DC Converter in DC Microgrid Applications Using Linear Matrix Inequality (LMI). 2019 IEEE Conference on Power Electronics and Renewable Energy (CPERE), IEEE, 2019, pp. 385–390.

[38] Dipayan Guha, Provas Kumar Roy, Subrata Banerjee, Load frequency control of interconnected power system using grey wolf optimization, Swarm Evol. Comput. 27 (2016) 97–115.

[39] Dipayan R8 Guha, Provas Kumar Roy, Subrata Banerjee, Load frequency control of large scale power system using quasi-oppositional grey wolf optimization algorithm, Eng. Sci. Technol., Int. J. 19 (4) (2016) 1693–1713.

[40] Feng Zheng, Qing-Guo Wang, Tong Heng Lee, On the design of multivariable PID controllers via LMI approach, Automatica 38 (3) (2002) 517–526.